

Zurich ServiceNow AI Platform Capabilities

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CMDB CI Class Models

The CMDB contains base-system classes that store data about configuration items (CIs). The CMDB CI Class Models ServiceNow® Store app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and dependent relationships if applicable.

You can use the added classes like any other class. Applications such as Discovery and Service Mapping can use class extensions to populate CIs and to discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

Related ServiceNow® Store apps and reference information:

- [Configuration Management database](#): A collection of class diagrams and class attributes for key CMDB classes.
- [CMDB tables descriptions](#): Descriptions of key CMDB tables in the base system.
- [Populating the CMDB](#): Information about the various options for populating the CMDB.
- [Discovery patterns](#): A ServiceNow Store app that provides a library of Discovery patterns for discovering specific devices and applications in the industry.
- [Service Graph Connectors](#): ServiceNow Store apps that provide predefined integrations for importing and integrating common third-party data into CMDB classes. Also includes the [IntegrationHub ETL](#) wizard for creating new ETL transform maps.

Add class models

The app adds classes, columns, and associated metadata as related records in the following tables:

- CMDB Class Information [cmdb_class_info]: Class descriptions
- Identifier [cmdb_identifier]: Identification rules
- Identifier Entry [cmdb_identifier_entry]: Identification entries

- CMDB Metadata Hosting Rules [cmdb_metadata_hosting]: Dependent relationships

Discover using extension classes

The following table lists the software and technologies that applications can discover using the extension classes. It provides links to documentation for CMDB CI Class Models and the corresponding discovery patterns.

Software/Technology	CMDB CI Class Models Store app	Discovery patterns
Avi load balancer	Avi load balancer extension classes	Avi Vantage load balancer discovery
BYOL Model of RDS for Oracle	BYOL model of RDS for Oracle extension classes	
Firewall	Firewall extension classes	
IBM Hardware Management Console (HMC)	IBM Hardware Management Console (HMC) extension classes	IBM Virtualization and Hardware Management Console discovery
Internet of Things (IoT)	Internet of Things (IoT) extension classes	N/A
Nutanix	Nutanix extension classes	Nutanix Acropolis discovery
OpenStack	OpenStack extension classes	OpenStack resource discovery
Red Hat Virtualization (RHV)	Red Hat Virtualization (RHV) extension classes	Red Hat Virtualization discovery

Software/Technology	CMDB CI Class Models Store app	Discovery patterns
Transport Layer Security (TLS)	Transport Layer Security (TLS) extension classes	Discovery procedures provided by Certificate Inventory and Management ServiceNow Store app
VMware NSX load balancer	VMware NSX load balancer extension classes	VMware NSX Advanced load balancer discovery

Request apps on the Store

Visit the [ServiceNow Store](#) website to view all the available apps and for information about submitting requests to the store. For cumulative release notes information for all released apps, see the [ServiceNow Store version history release notes](#).

Verify successful installation

After installing the CMDB CI Class Models store app, make sure the classes were added successfully:

1. Navigate to **All > Configuration > CI Class Manager**.
2. Click **Hierarchy** to display the CI Classes list.

This list contains the added classes, such as the Nutanix classes.

3. Select a class to see the corresponding class details, identification rules, identifier entries, and dependent relationships, if applicable.

Warning: Uninstalling the CMDB CI Class Models application might compromise the integrity of the CMDB and result in unexpected behavior.

API extension classes

The CMDB CI Class Models ServiceNow Store app adds or updates classes for APIs (application programming interface).

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

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APIs

APIs are a set of definitions and protocols that enable computer programs to communicate with each other, which enables you to build or integrate application software. APIs typically use web-based technology to communicate with other APIs. APIs are generally used to programmatically perform jobs or tasks, or to view, import, export, delete, or modify data.

The classes added in this release extend the CMDB data model and provide a foundation for the representation of API CI classes. You can use this foundation to do the following:

- Gain greater visibility into your APIs.
- Identify security issues and vulnerabilities associated with an API endpoint.

Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. For additional detail, see the [Community article titled "New Data Model in CMDB for APIs"](#).

CMDB CI Class Models: The following classes for API are available. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Extends	Description
API [cmdb_ci_api]	Configuration Item [cmdb_ci]	APIs that enable two computer programs to communicate with each other, typically using web-based technologies. Example: ChatAPI (https://[apiID].execute-api.use-east-2.amazonaws.com).
API Component [cmdb_ci_api_component]	Configuration Item [cmdb_ci]	Reusable objects related to your API definition that facilitate functionality or exchange of data. API Headers related list. Example: GET https://[instance].servicenow.com/api/now/table/{tablename}.
API Frontend [cmdb_ci_api_frontend]	API Component [cmdb_ci_api_component]	The part of an API from which a client or user interacts or makes requests. Example: GET https://[apiID].execute-api.us-east-2.amazonaws.com/{proxy+}.
API Backend [cmdb_ci_api_backend]	API Component [cmdb_ci_api_component]	The part of an API that fulfills requests by interacting with backend services, such as servers. Example:

Class	Extends	Description
		Lambda:Chat-API-Proxy.
API Gateway [cmdb_ci_api_gateway]	Application [cmdb_ci_appl]	API infrastructure that centralizes client API requests and manages backend processes and services. Example: Kong Gateway.
Managed API [cmdb_ci_managed_api]	Configuration Item [cmdb_ci_api]	API discovered from a gateway or management service. You can enforce a dependency on a gateway for APIs in this class.

Class attributes

CMDB CI Class Models: Release 1.49.0 adds the following attributes to the respective classes.

API [cmdb_ci_api]

Attribute	Data type	Description
Base URL	String (1024)	Base address from which all API components extend.
ID	String (1024)	Unique identifier from the source system.
Type	Choice list	Type of API. You can specify: • REST • SOAP

Attribute	Data type	Description
		• HTTP • gRPC • GraphQL • Websocket
Version	Numeric	Version of the API.
Spec Location	URL	URL to the location of the API specification. Example: OpenAPI spec definition.
Technical Specification Location	String	Location of the technical specification repository for an API. Example: GitHub repository containing the deployed API code.
Design Specification Location	String	URL to the location of the API design or documentation. Example: SwaggerHub API Design Specification URL.

The API [cmdb_ci_api] class supports mapping of tags and labels to the Key Value [cmdb_key_value] table to enable key-value storage for API metadata.

API Component [cmdb_ci_api_component]

Attribute	Data type	Description
Method	String	REST API methods. Examples: • GET

Attribute	Data type	Description
		• POST • DELETE
Protocol	String	Communication protocol. Example: HTTP, HTTPS.
Host	String (100)	System that hosts the API.
Path	String (1024)	Specific route the API follows.
Port	String	Communication port. Example: 80, 443, and so on.
URL	String (1024)	URL of the resource being called.
ID	String (1024)	Unique identifier from the source system.
Internet Facing	Boolean	Boolean that denotes whether the component is reachable from the public internet. Specify 1 or "true" if the component is reachable.
Authorization	String	Type of authorization or authentication method. Example: • Basic • Key

Attribute	Data type	Description
		• OAuth • None
Request data types	String (255)	List of data types in the request. Examples: • CC • Email • Address
Response data types	String (255)	List of data types in the response. Examples: • CC • Email • Address

The API Component [cmdb_ci_api_component] class supports mapping of tags and labels to the Key Value [cmdb_key_value] table to enable key-value storage for API component metadata.

API Frontend [cmdb_ci_api_frontend]

Attribute	Data type	Description
Parent ID	Reference to [cmdb_ci_api_frontend]	Reference to a parent API component.

API Backend [cmdb_ci_api_backend]

Attribute	Data type	Description
Type	String	Backend protocol types of the API. Examples: • Lambda • HTTP • Logic App

API Gateway [cmdb_ci_api_gateway]

Attribute	Data type	Description
ID	String (255)	Unique identifier from the source system.

The API Gateway [cmdb_ci_api_gateway] class supports mapping of tags and labels to the Key Value [cmdb_key_value] table. The mapping enables key-value storage for API gateway metadata. The following marker classes are available for API gateways. These classes inherit all attributes from the API Gateway [cmdb_ci_api_gateway] parent class.

- Anypoint API Gateway [cmdb_ci_api_gateway_anypoint] (provided by MuleSoft)
- Tyk API Gateway [cmdb_ci_api_gateway_tyk] (provided by Tyk)
- Boomi API Gateway [cmdb_ci_api_gateway_boomi] (provided by Boomi)

The marker classes follow the existing class structure and do not require changes to Identification and Reconciliation engine (IRE) rules.

Note: The Managed API [cmdb_ci_managed_api] class is specific to APIs discovered from gateways and other managed services, and does not introduce new attributes.

In API gateways and developer portals, an API product bundle is a collection of one or more APIs for consumption by developers and applications. Some platforms use the term for an API product alone.

API Product Bundle [cmdb_ci_api_product_bundle]

Attribute	Data type	Description
ID	String (255)	Unique identifier of the API product bundle.
Discovered Approval Type	String	Type of request approval as discovered from the source system.
Discovered Access Type	String	Access configuration as discovered from the source system. For example, public or internal.
Creation Date	Date/Time	Date and time when the API product bundle was initially discovered or recorded.
Last Modified Date	Date/Time	Date and time of the most recent update to the API product bundle.
Discovered State	String	Current state of the product bundle as reported by the source. For example, published or deprecated.

To access one or more API product bundles, a developer can register for an API consumer subscription on a developer portal. The subscription registers with one or more bundles and provides a key for access to the APIs in the bundles.

API Consumer Subscription [cmdb_ci_api_consumer_subscription]

Attribute	Data type	Description
Discovered Scope	String	Scope of access as discovered from the source system. For example, all APIs or specific APIs.
API Consumer	Reference	Reference to the API consumer record. For example, cmdb_ci_api_consumer.
ID	String (255)	Unique identifier of the API consumer subscription.
Discovered State	String	Subscription status as discovered from the source system. For example, active or inactive.
Creation Date	Date/Time	Date and time when the API consumer subscription was created in the source system.
Last Modified Date	Date/Time	Date and time of the most recent update to the API consumer subscription.

The WebACL class [cmdb_ci_web_acl] represents ACLs for CloudFront, API gateway REST APIs, application load balancers, AppSync GraphQL APIs, Cognito user pools, App Runner services, AWS Verified Access, and the Azure Front Door application gateway.

Key relationship structures

There are a number of key relationships that must be defined for API and Kong classes.

API relationships

Parent class	Relationship	Child class	Relationship type
API [cmdb_ci_api]	Uses::Used by	API Component [cmdb_ci_api_component]	Suggested
API Gateway [cmdb_ci_api_gateway]	Provides::Provided By	Managed API [cmdb_ci_managed_api]	Dependent
API Frontend [cmdb_ci_api_frontend]	Use End Point To::Use End Point From	API Backend [cmdb_ci_api_backend]	Suggested
API Backend [cmdb_ci_api_backend]	Uses::Used By	Kong Load Balancer [cmdb_ci_kong_lb]	Suggested
API Gateway [cmdb_ci_api_gateway]	Provides::Provided By	API Product Bundle [cmdb_ci_api_product_bundle]	Dependent
API Gateway	Provides::Provided By	API Consumer Subscription	Dependent

Parent class	Relationship	Child class	Relationship type
[cmdb_ci_api_gateway]		[cmdb_ci_api_consumer_subscription]	
API Product Bundle	Contains::Contained by	API [cmdb_ci_api]	Suggested
API Product Bundle	Used by::Uses	API Consumer Subscription [cmdb_ci_api_consumer_subscription]	Suggested

Related non-CMDB tables

CMDB CI Class Models v 1.49.0 introduces the following non-CMDB tables as related lists for the following API extension classes:

API related list

API Deployment [api_deployment]

Attribute	Data type	Description
Name	String (100)	Name of the API deployment.
API	Reference	Reference to the deployed API (cmdb_ci_api).
Unmatched API Endpoint	Reference	Reference to the unmatched API endpoint, if the API doesn't match an existing API

Attribute	Data type	Description
		(cmdb_ci_unmatched_api_endpoint).
Configuration Item	Reference	Reference to the Configuration Item. This is typically manually specified as a reference, if you know what CI the API is deployed to.

Note: The API Deployment non-CMDB table relates to both the API [cmdb_ci_api] and Unmatched API Endpoint [cmdb_ci_unmatched_api_endpoint] classes.

API Component related list

API Header [api_header]

Attribute	Data type	Description
Name	String (100)	Name of the API header.
API Component	Reference	Reference to the component where the API header is defined (cmdb_ci_api_component).
Unmatched API Endpoint	Reference	Reference to the unmatched API when the endpoint can't be matched to an existing API or component (cmdb_ci_unmatched_endpoint).

API Gateway related lists

API Consumer [api_consumer]

Attribute	Data type	Description
Username	String (100)	Name of the API consumer. Note: The Consumer field in the api_policy table is a reference field. To display the username in the Username field on the API Policies list, you must set the display attribute of Username to <code>true</code>. If the attribute is not set to <code>true</code>, then the list displays the Sys ID instead.
ID	String (255)	Unique identifier from the source system.
Custom ID	String (100)	Alternate display name of the user.
API Gateway	Reference	Reference to the gateway where the consumer is defined (cmdb_ci_api_gateway).

Attribute	Data type	Description
API Consumer Type	Choice list	API user type (user_type). You can specify: • Internal • External • Unknown
Discovered State	String	Represents the current status of the API consumer's account.
Provider	String	Name of the identity management provider responsible for authenticating the API consumer.
Email	String	Email address of the API consumer.
Registration Date	Date/Time	Date the consumer registered with the providing platform.

API Consumer Access [api_consumer_access]

Attribute	Data type	Description
API Consumer	Reference	Reference to the API consumer record. For example, cmdb_ci_api_consumer.
API	Reference	Reference to the API [cmdb_ci_api] to

Attribute	Data type	Description
		which the consumer is requesting access.
Auth Method	String	Authentication method used by the API consumer to access the API. For example, API key or OAuth.
State	Choice	Indicates whether the access is currently active or inactive.
Last used	Date/Time	Last date and time the API consumer accessed the API.
Valid to	Date/Time	Expiration date of the consumer's access to the API.

API Policy [api_policy]

Attribute	Data type	Description
Name	String (100)	Name of the API policy.
ID	String (255)	Unique identifier from the source system.
Frontend	Reference	Reference to the API Frontend (cmdb_ci_api_frontend).
Managed API	Reference	Reference to the Managed API

Attribute	Data type	Description
		(cmdb_ci_managed_api).
Consumer	Reference	Reference to the API Consumer (api_consumer) non-CMDB table. Note: Consumer is a reference field. To display the api_consumer (Username) in the API Policies list, you must set the display attribute to <code>true</code> for Username field on the api_consumer table. If the attribute is not set to <code>true</code>, then the list displays the Sys ID instead.
Protocols	String	Array of protocols that this API policy can apply to.
Active	Boolean	Determines if this non-CMDB table is considered active or inactive.
API Gateway	Reference	Reference to the gateway where the consumer is defined (cmdb_ci_api_gateway).

Unmatched API extension classes

The CMDB CI Class Models store app adds or updates classes for unmatched APIs. API endpoints that are not structured well enough to populate the API [cmdb_ci_api] and API Component [cmdb_ci_api_component] classes instead populate the Unmatched API Endpoint [cmdb_ci_unmatched_api_endpoint] class. Unmatched APIs are typically used by integrations like security scanners or observability tools.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

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Unmatched APIs

API endpoints that are not structured well enough to populate the API [cmdb_ci_api] and API Component [cmdb_ci_api_component] classes instead populate the Unmatched API Endpoint [cmdb_ci_unmatched_api_endpoint] class. Unmatched APIs are typically used by integrations like security scanners or observability tools.

Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates.

CMDB CI Class Models: Release 1.49.0 adds the following class for unmatched API endpoints. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Extends	Description
Unmatched API Endpoint [cmdb_ci_unmatched_api_endpoint]	Configuration Item [cmdb_ci]	APIs with unstructured endpoints that cannot populate cmdb_ci_api or cmdb_ci_api_component.

Class attributes

CMDB CI Class Models: Release 1.49.0 adds the following attributes to the Unmatched API Endpoint [cmdb_ci_unmatched_api_endpoint] class.

Unmatched API Endpoint [cmdb_ci_unmatched_api_endpoint]

Attribute	Data type	Description
Method	String	Backend protocol types of the API. Examples: • Lambda • HTTP • Logic App
Protocol	String	Communication protocol. Example: HTTP, HTTPS.
URL	String (1024)	URL of the resource being called.
ID	String (1024)	Unique identifier from the source system.
Internet Facing	Boolean	Boolean that denotes whether the component is

Attribute	Data type	Description
		reachable from the public internet. Specify 1 or "true" if the component is reachable.
Authorization	String	Type of authorization or authentication method. Example: • Basic • Key • Oauth • None
Request data types	String (255)	List of data types in the request. Examples: • CC • Email • Address
Response data types	String (255)	List of data types in the response. Examples: • CC • Email • Address

Key Relationship Structures

The Unmatched API Endpoint [cmdb_ci_unmatched_api_endpoint] CMDb class and API Endpoint Discovered [api_endpoint_discovered] non-CMDb table are intended for scenarios where unstructured API data is ingested and needs to be processed beyond the capabilities of IntegrationHub-ETL.

API Endpoint Discovered [api_endpoint_discovered] can serve as a staging table for raw data for the API [cmdb_ci_api] and API Component [cmdb_ci_api_component] classes. API data that does not come from a well-structured data source or cannot have standard identification rules applied can first populate the API Endpoint Discovered [api_endpoint_discovered] table. You can then use a customer-defined integration to ensure that well-formed data that matches the quality and condition requirements for API [cmdb_ci_api] and API Component [cmdb_ci_api_component] populates the most appropriate table.

In cases where the data cannot be correctly parsed or is structured in an unexpected fashion, you can instead use a customer-defined integration to populate the Unmatched API Endpoint [cmdb_ci_unmatched_api_endpoint] table.

Data that populates the API Endpoint Discovered [api_endpoint_discovered] table is purged every 30 days.

API Endpoint Discovered [api_endpoint_discovered]

Attribute	Data type	Description
Name	String (100)	Name of the API endpoint.
Method	String	REST API methods. Examples: • GET • POST • DELETE

Attribute	Data type	Description
URL	String (1024)	URL of the resource being called.
Type	Choice list	Type of API. You can specify: • REST • SOAP • HTTP • gRPC • GraphQL • Websocket
Authorization	String	Type of authorization or authentication method. Example: • Basic • Key • OAuth • None
Internet Facing	Boolean	Boolean that denotes whether the component is reachable from the public internet. Specify 1 or "true" if the component is reachable.
Request Datatypes	String (255)	List of data types in the request. Examples:

Attribute	Data type	Description
		• CC • Email • Address
Response Datatypes	String (255)	List of data types in the response. Examples: • CC • Email • Address
Headers	String (255)	Comma-separated list of header names.
Configuration Item	Reference	Reference to a matching CMDB CI (API Component or Unmatched API Endpoint).

Related non-CMDB tables

The Unmatched API extension class uses the API Deployment [api_deployment] non-CMDB table as a related list:

API Deployment [api_deployment]

Attribute	Data type	Description
Name	String (100)	Name of the API deployment.

Attribute	Data type	Description
API	Reference	Reference to the deployed API (cmdb_ci_api).
Unmatched API Endpoint	Reference	Reference to the unmatched API endpoint, if the API doesn't match an existing API (cmdb_ci_unmatched_api_endpoint).
Configuration Item	Reference	Reference to the Configuration Item. This is typically manually specified as a reference, if you know what CI the API is deployed to.

Avi load balancer extension classes

The CMDB CI Class Models store app adds or updates classes for the Avi load balancer.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

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Request apps on the Store

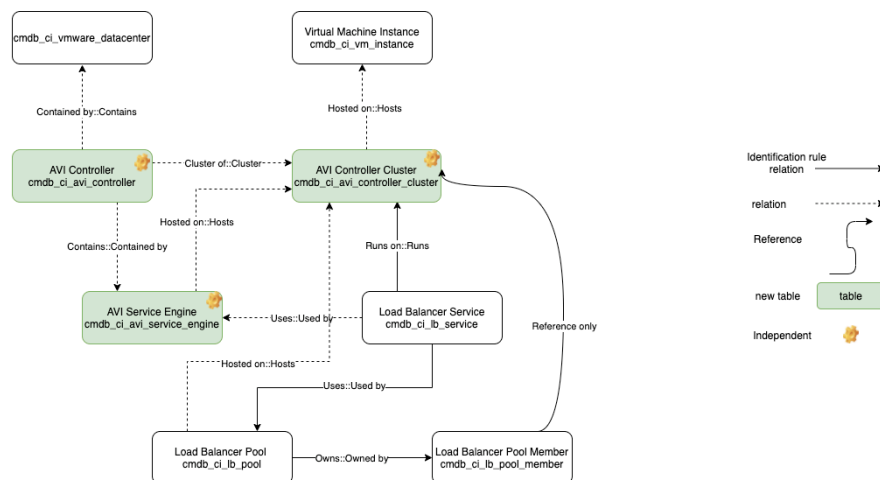
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Avi load balancer

The Avi Vantage platform is built on software-defined principles, enabling a next-generation architecture to deliver the flexibility and simplicity expected by IT and lines of business. The Avi Vantage platform architecture separates the data and control planes to deliver application services beyond load balancing, such as application analytics, predictive autoscaling, micro-segmentation, and self-service for app owners in on-premises or cloud environments. The platform provides a centrally managed, dynamic pool of load balancing resources on commodity x86 servers, virtual machines, or containers, to deliver granular services close to individual applications. Providing these services allows network services to scale near infinitely without the added complexity of managing hundreds of disparate appliances.

Discovery uses the [Avi Vantage load balancer discovery](#) pattern to find Avi load balancer resources.

Avi load balancer classes integrated with the CMDB class hierarchy



Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. CMDB CI Class Models: Release 1.6.0 adds the following classes for the Avi load balancer. For the list of classes in the base

system, including classes that this app might extend, see [Cmdb tables descriptions](#).

Class	Extends	Description
Avi Controller [cmdb_ci_avi_controller]	Virtual Machine Object [cmdb_ci_vm_object]	Avi Controller is a single point of management and control that is the 'brain' of the entire Avi Vantage system, and typically deployed as a redundant three-node cluster.
Avi Controller Cluster [cmdb_ci_avi_controller_cluster]	Virtual Machine Object [cmdb_ci_vm_object]	Avi Controller cluster uses big data analytics to analyze the data and present actionable insights to administrators on intuitive dashboards on the Avi Admin Console.
Avi Service Engine [cmdb_ci_avi_service_engine]	Virtual Machine Object [cmdb_ci_vm_object]	Avi Service Engines (SEs) handle all data plane operations within Avi Vantage by receiving and executing instructions from the Avi Controller.

Class columns

Cmdb CI Class Models: Release 1.6.0 adds the following column to the respective class.

Avi Service Engine [cmdb_ci_avi_service_engine]

Added column	Description
version	The version of the Avi Service Engine resource.

Related concepts

- [CMDB schema model](#)

BYOL model of RDS for Oracle extension classes

The CMDB CI Class Models store app adds or updates classes for the BYOL Model of RDS for Oracle.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

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BYOL Model of RDS for Oracle

The Amazon RDS for Oracle is a fully managed commercial database that makes it easy to set up, operate, and scale Oracle deployments in the cloud. You can run Amazon RDS for Oracle in one of two different licensing models: "License Included" and "Bring-Your-Own-License (BYOL)". In the "License Included" service model, you do not need separately purchased Oracle licenses; the Oracle Database software has been licensed by AWS.

The base system supports discovery of the RDS Databases and their licenses.

Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. CMDB CI Class Models: Release 1.23.0 adds the following classes for the BYOL Model of RDS for Oracle. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Description
cmdb_ci_cloud_database	The cloud databases.
cmdb_ci_serverless_hardware	Hardware type information of the databases.

Class Columns

CMDB CI Class Models: Release 1.23.0 adds the following column to the respective class.

Serverless Hardware [cmdb_ci_serverless_hardware]

Added column	Description
cloud_vendor	The cloud vendor.
host_type	The host type such as PaaS/IaaS.
cpu_core_count	Amount of CPU cores.
cpu_core_thread	Amount of CPU threads.
cpu_count	Amount of CPUs.
object_id	ID of the CI.

Cloud Database [cmdb_ci_cloud_database]

Added column	Description
multi_az	Determines if the database is deployed on multiple availability zones (true/false).
replication_enabled	Determines if replication is enabled (true/false).
replication_type	Replication type.
replica_source	Database name of the replication source.

Cassandra extension classes

The CMDB CI Class Models store app adds or updates classes for Cassandra databases.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

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Cassandra

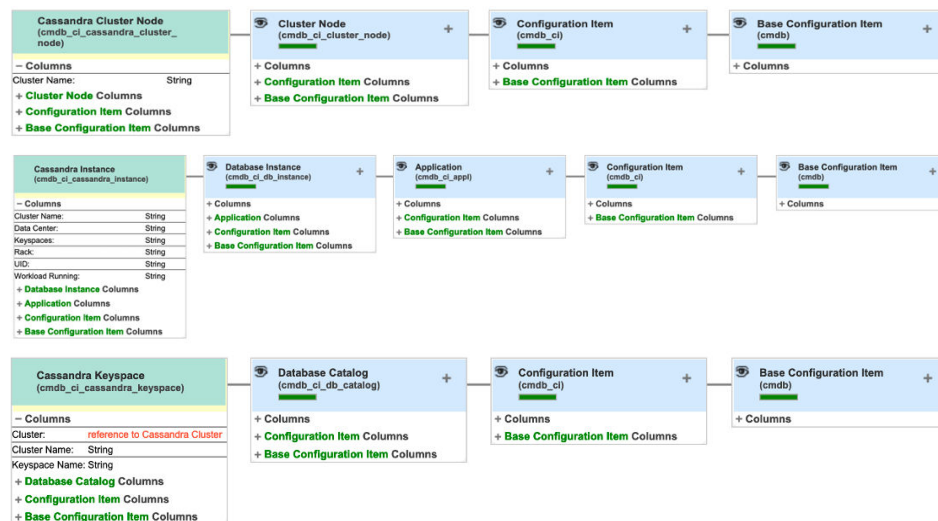
Cassandra is a distributed database that is treated as one database and that runs on a cluster of Linux nodes. A Cassandra cluster CI represents a logical entity that doesn't refer to a Linux node.

Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Cassandra classes

Class	Extends	Description
Cassandra cluster node [cmdb_ci_cassandra_cluster_node]	Cluster node [cmdb_ci_cluster_node]	• "Hosts" cmdb_ci_cassandra_instance • "Cluster Of" cmdb_ci_cassandra_cluster • "Hosted On" cmdb_ci_server
Cassandra instance [cmdb_ci_cassandra_instance]	DB instance [cmdb_ci_db_instance]	• "Runs On" cmdb_ci_server • "Hosted on" cmdb_ci_cassandra_cluster_node • "Contains" cmdb_ci_config_file_tracked
Cassandra keyspace [cmdb_ci_cassandra_keyspace]	DB catalog [cmdb_ci_db_catalog]	"Hosted On" cmdb_ci_cassandra_cluster



DNS resource record classes

The CMDB CI Class Models ServiceNow Store app adds or updates classes for DNS records, also known as a DNS resource records (RR).

Request apps on the Store

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DNS zone resource records

A DNS record is an entry in the domain name system (DNS) database that provides information about a specific aspect of a domain name. Each DNS record contains data associated with a domain name, such as IP addresses, mail server addresses, or other resources. Each type of record serves a specific purpose, such as mapping domain names to IP addresses (A records), identifying mail servers for a domain (MX records), and providing aliases for domain names (CNAME records), among others. DNS resource records are essential for the proper functioning of the DNS infrastructure on the internet.

See the [full release notes for all CMDB CI Class Models](#).

Note: In CMDB CI Class Models release 1.57, existing classes in the CMDB schema that represent DNS resource records like A, AAAA, CName, and so on are deprecated. The following classes are defined with an ideal base, child class, and identification rule.

Supported DNS resource record types

- A Record: Holds the IP address of a domain.
- AAAA Record: Contains the IPv6 address of a domain.
- CNAME Record: Forwards one domain or subdomain to another domain. Does not provide an IP address.
- MX Record: Directs mail to an email server.

The supported record types enable actions like the following:

- Capture domain name registration in the CMDB and create workflow to request, approve, and track life cycle.
- Auto-populate the data from integration to the registrar.
- Connect the data to an IPAM record.
- Connect the data to the certificate on the DNS.
- Connect the data to the CDN provider.

DNS resource record classes integrated with the CMDB class hierarchy

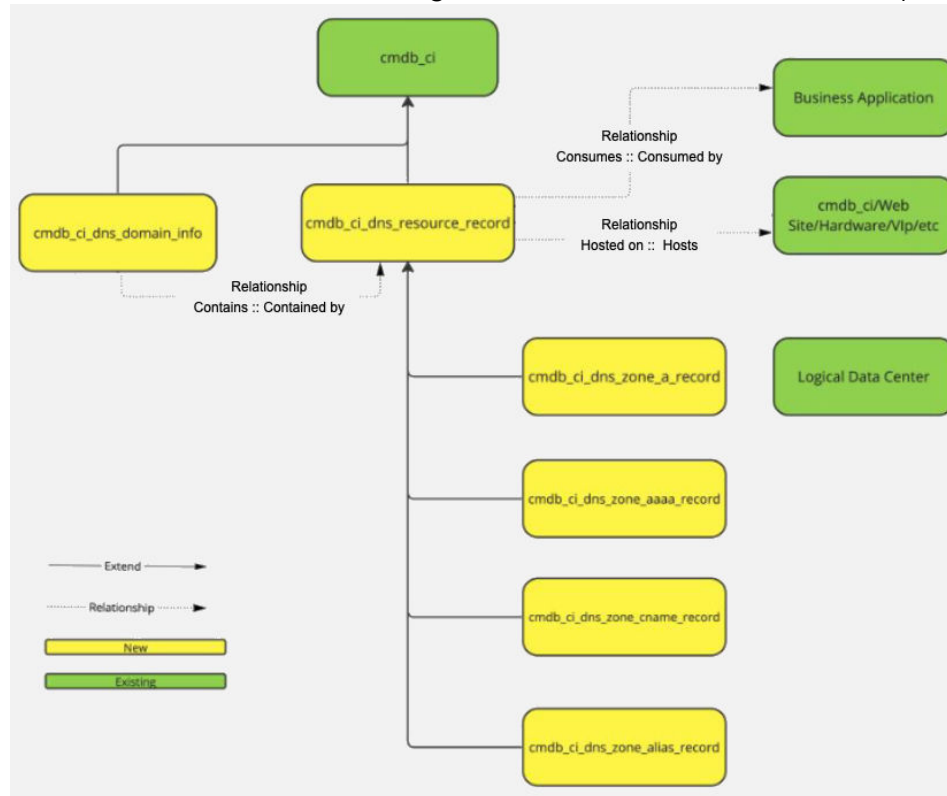


Table: cmdb_ci_dns_resource_record

This section lists the classes that the CMDB CI Class Models store app adds or updates. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

DNS record detail from Domain Name Server (DNS) for a WWW domain.

Column	Description	Details
DNS Service Provider [service_provider]	Defines which DNS service provider supplied this value, examples like Cloudflare, Google	Identification Rule (name+service_provider) allow null

Column	Description	Details
Replaces the deprecated column: DNS Service Provider [name]	DNS, OpenDNS, Local DNS, Private DNS, empty, and so on.	String (255)
Time to Live [ttl]	Time, in seconds, that a DNS record will be cached by an outside DNS server or resolver.	integer
Record Class [record_class]	Class of DNS record.	• IN (Internet) (default) • CH (Chaosnet) • HS (Hesiod)

Table: cmdb_ci_dns_zone_a_record

A records hold only IPv4 addresses.

Column	Description	Details
A IP Address [ip_address]	IPv4 address	ip_addr Identification Rule (name+service_provider+ip_address) allow null

Table: cmdb_ci_dns_zone_aaaa_record

AAAA records hold only IPv6 addresses.

Column	Description	Details
AAAA IP Address [ip_address]	IPv6 address	ip_addr Identification Rule (name+service_provider+ip_address) allow null

Table: cmdb_ci_dns_zone_cname_record

A canonical name (CNAME) record points from an alias domain to a canonical domain.

Column	Description	Details
Canonical Name [cname]	Fully qualified domain name / Canonical Name.	String (255) Identification Rule (name+service_provider+cname) allow null

Table: cmdb_ci_dns_zone_alias_record

An ALIAS record points domain name to a hostname instead of an IP address.

Column	Description	Details
Alias Name [alias_name]	Alias name of the DNS record.	String (255) Identification Rule (name+service_provider+alias_name) allow null

Table: cmdb_ci_dns_domain_info

Domain Name enables you to trace ownership and tenure of domain name details using DNS lookup tools like whois.

Column	Description	Details
Domain Name [domain_name]	WWW Domain Name value from the DNS lookup tool.	String (255) Identification Rule (Name)
Registry Domain ID [registry_domain_id]	Registry Domain ID value from the DNS lookup tool.	String (255)
Registrar [registrar]	Registrar value from the DNS lookup tool.	String (255)
Domain Creation Date [domain_creation_date]	Creation Date of the domain from the DNS lookup tool.	glide_date_time
Domain Updated Date [domain_updated_date]	Updated Date of the domain from the DNS lookup tool.	glide_date_time
Registrar Registration Expiration Date [registrar_registration_expiration_date]	Registrar Registration Expiration Date value from the DNS lookup tool.	glide_date_time

Extend classes and rules for third-party integrations

Extend and update CMDB CI Class Models store app classes and rules when using third-party integration tools.

When using a third-party tool to integrate with ServiceNow apps, gaps can occur between the integration and different CIs. Some of the

integrated tables and classes may be missing classes and rules that are not included with the CMDB CI Class Models store app.

See the [full release notes for all CMDB CI Class Models](#).

This section lists the classes that the CMDB CI Class Models store app adds or updates. The classes and rules in the following table enable you to add and extend the CMDB CI Class Models store app for integrations.

Table	Extends	Rules and related entries
Postgresql Schema cmdb_ci_postgresql_schema	cmdb_ci_db_catalog	Containment rule: cmdb_ci_db_postgresql_instance- >Contains:Contained By- >cmdb_ci_postgresql_schema
		Identification Rule: Dependent, Attributes: name
		Related Entries (cmdb_related_entry): - Identifier: cmdb_ci_postgresql_schema - Related table: cmdb_key_value - Referenced field: configuration_item

Table	Extends	Rules and related entries
Information Object cmdb_ci_information_object	cmdb_ci	Identification rule: Independent, attributes: name Related Entries (cmdb_related_entry): - Identifier: cmdb_ci_information_object - Related table: cmdb_key_value - Referenced field: configuration_item
Oracle Catalog cmdb_ci_db_ora_catalog	cmdb_ci_db_catalog	Related Entries (cmdb_related_entry): - Identifier: cmdb_ci_db_ora_catalog - Related table: cmdb_key_value - Referenced field: configuration_item
MySQL Catalog cmdb_ci_db_mysql_catalog	cmdb_ci_db_catalog	Containment rule: cmdb_ci_db_mysql_instance- >Contains:Contained By-

Table	Extends	Rules and related entries
		>cmdb_ci_db_mysql_catalog Identification rule: Dependent, attributes: name
		Related Entries (cmdb_related_entry): - Identifier: cmdb_ci_db_mysql_catalog - Related table: cmdb_key_value - Referenced field: configuration_item
MS SQL Database cmdb_ci_db_mssql_database	cmdb_ci_db_instance	Related Entries (cmdb_related_entry): - Identifier: cmdb_ci_db_mssql_database - Related table: cmdb_key_value - Referenced field: configuration_item
Sybase Catalog	cmdb_ci_db_catalog	Containment rule:

Table	Extends	Rules and related entries
cmdb_ci_db_syb_catalog		cmdb_ci_db_syb_instance- >Contains:Contained By- >cmdb_ci_db_syb_catalog
		Identification rule: Dependent, attributes: name
		Related Entries (cmdb_related_entry): - Identifier: cmdb_ci_db_syb_catalog - Related table: cmdb_key_value - Referenced field: configuration_item
DB2 Database (formerly DB2 Catalog) cmdb_ci_db_db2_catalog	cmdb_ci_db_catalog	Containment rule: cmdb_ci_db_db2_instance- >Contains:Contained By- >cmdb_ci_db_db2_catalog
		Identification rule:

Table	Extends	Rules and related entries
		Dependent, attributes: name Related Entries (cmdb_related_entry): - Identifier: cmdb_ci_db_db2_c atalog - Related table: cmdb_key_value - Referenced field: configuration_item

For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Related concepts

- [CMDB schema model](#)

Firewall extension classes

The CMDB CI Class Models store app adds or updates classes for firewall devices.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

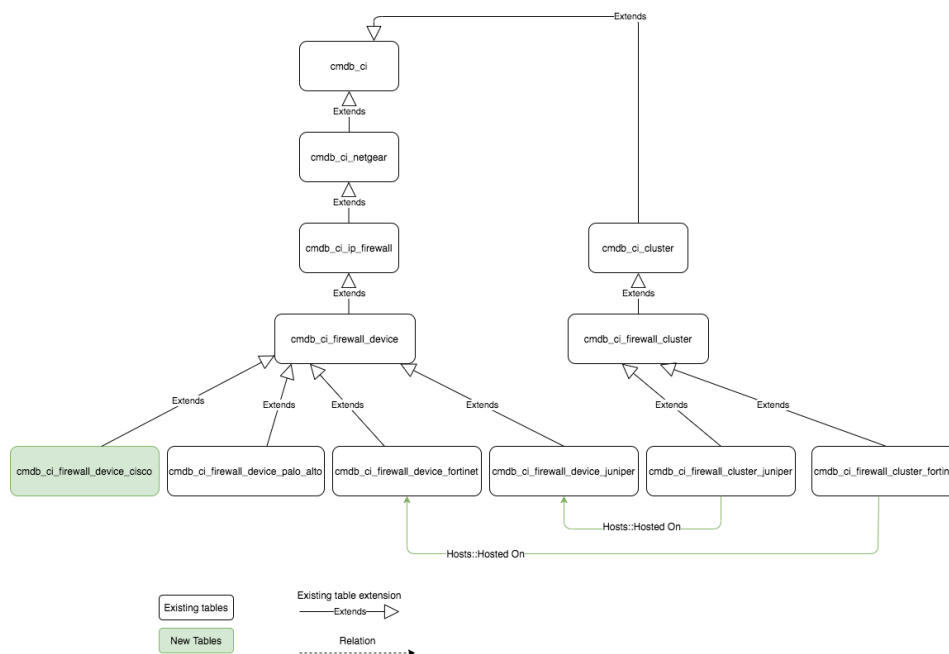
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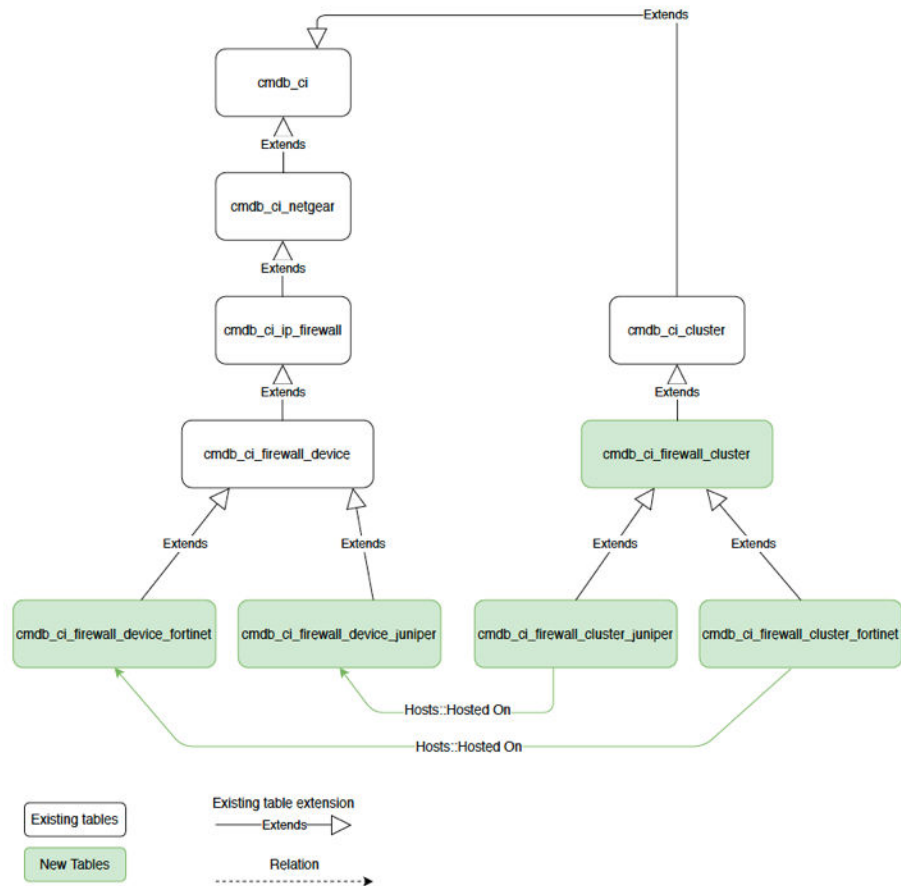
Firewalls

A firewall is a network security system that, based on security policies, monitors and controls incoming and outgoing network traffic. Firewalls typically form a barrier between an internal network and an untrusted external network such as the internet. It usually consists of security policies that help secure an organization from external threats and cyberattacks. Firewall vendors may provide a centralized firewall manager to manage many firewall devices and the security policies residing on them. For example, Panorama™ is the centralized management system for Palo Alto Networks firewalls.

Firewall extension classes integrated with the CMDB class hierarchy (CMDB CI Class Models: Release 1.11.0)



Firewall extension classes integrated with the CMDB class hierarchy
(CMDB CI Class Models: Release 1.10.0)



Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. See the class columns table for further details about the columns added for each class. CMDB CI Class Models: Release 1.10.0 adds or updates the following classes for the discovery of network firewall devices. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Extends	Description
IP Firewall [cmdb_ci_ip_firewall]	NETGEAR [cmdb_ci_netgear]	Contains all network firewalls.
Firewall Device [cmdb_ci_firewall_device]	IP Firewall [cmdb_ci_ip_firewall]	Network security system that monitors and controls incoming and outgoing network traffic, based on security policies.
Fortinet Firewall Device [cmdb_ci_firewall_device_fortinet]	Firewall Device [cmdb_ci_firewall_device]	Fortinet firewall device.
Fortinet Firewall Interface [cmdb_ci_fortinet_firewall_interface]	Network Interface [cmdb_ci_ni_interface]	Discovered by the Next Generation Fortinet Network Firewall - REST pattern. See Fortinet firewall and FortiGate VDOM REST-based discovery .
Fortinet Firewall Policy [cmdb_ci_fortinet_firewall_policy]	Firewall Security Policy [cmdb_ci_firewall_security_policy]	Discovered by the Next Generation Fortinet Network Firewall - REST pattern. See Fortinet firewall and FortiGate VDOM REST-based discovery .
Fortinet Virtual Domain [cmdb_ci_fortinet_vdom]	IP Firewall [cmdb_ci_ip_firewall]	Discovered by the Next Generation Fortinet Network Firewall - REST pattern. See Fortinet firewall

Class	Extends	Description
		and FortiGate VDOM REST-based discovery.
Juniper Firewall Device [cmdb_ci_firewall_device_juniper]	Firewall Device [cmdb_ci_firewall_device]	Juniper firewall device.
Firewall Device Group [cmdb_ci_firewall_device_group]	CMDB CI [cmdb_ci]	Group of firewall devices.
Panorama Firewall Device Group [cmdb_ci_firewall_device_group_panorama]	Firewall Device Group [cmdb_ci_firewall_device_group]	Group of Panorama firewall devices.
Palo Alto Firewall Device [cmdb_ci_firewall_device_palo_alto]	Firewall Device [cmdb_ci_firewall_device]	Palo Alto firewall device.
Firewall Cluster [cmdb_ci_firewall_cluster]	[cmdb_ci_cluster]	Group of firewall nodes that work as a single logical entity.
Fortinet Firewall Cluster [cmdb_ci_firewall_cluster_fortinet]	Firewall Cluster [cmdb_ci_firewall_cluster]	Fortinet firewall cluster.
Juniper Firewall Cluster [cmdb_ci_firewall_cluster_juniper]	Firewall Cluster [cmdb_ci_firewall_cluster]	Juniper firewall cluster.
Firewall Manager [cmdb_ci_firewall_manager]	CMDB CI [cmdb_ci]	System that provides centralized management for many firewall devices and the security policies residing on them.

Class	Extends	Description
Panorama Firewall Manager [cmdb_ci_firewall_manager_panorama]	Firewall Manager [cmdb_ci_firewall_manager]	The centralized network security management tool for Palo Alto Networks firewalls.
Firewall Security Policy [cmdb_ci_firewall_security_policy]	CMDB CI [cmdb_ci]	The security policy that the firewall device enforces.
Panorama Firewall Security Policy [cmdb_ci_firewall_security_policy_panorama]	Firewall Security Policy [cmdb_ci_firewall_security_policy]	The security policy that the Panorama firewall device enforces.

CMDB CI Class Models: Release 1.12.0 adds the following class for the discovery of network firewall devices.

Class	Extends	Description
Cisco Firewall Device [cmdb_ci_firewall_device_cisco]	Firewall Device [cmdb_ci_firewall_device]	All Cisco Firewall devices.

Class columns

CMDB CI Class Models: Release 1.10.0 adds the following columns to the respective classes.

IP Firewall [cmdb_ci_ip_firewall] class

Added columns	Description
Hardware Operating System	OS running on the hardware.
Hardware OS Version	OS version running on the hardware.

Firewall Cluster [cmdb_ci_firewall_cluster] class

Added columns	Description
Hardware Operating System	OS running on the hardware.
Hardware OS Version	OS version running on the hardware.

CMDB CI Class Models: Release 1.12.0 adds no columns to the existing classes.

Related concepts

- [CMDB schema model](#)

IBM Hardware Management Console (HMC) extension classes

The CMDB CI Class Models store app adds or updates classes for the IBM Hardware Management Console (HMC).

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

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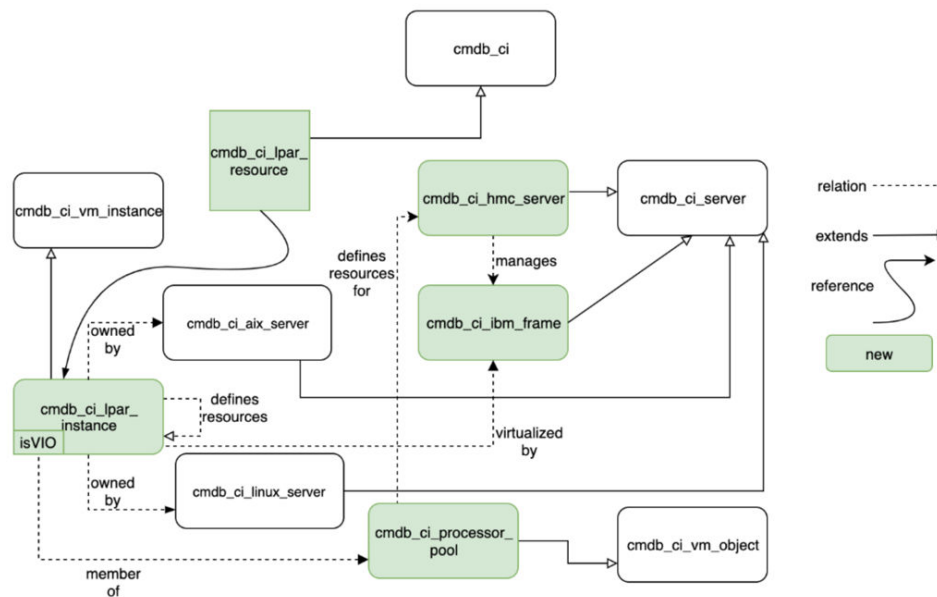
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IBM Hardware Management Console (HMC)

The IBM HMC extension classes support discovery of IBM virtualization technology by providing:

- Discovery of LPARs/Frames, which SAM use cases need
- Topology data of IBM HMC, which event correlation requires
- Topology data for ITSM use cases, such as in-frame migration

IBM HMC extension classes integrated with the CMDB class hierarchy



Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. See the class columns table for further details about the columns added for each class. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

CMDB CI Class Models: Release 1.3.0 adds the following classes for the IBM HMC.

Class	Extends	Description
IBM Frame [cmdb_ci_ibm_frame]	Server [cmdb_ci_server]	IBM physical machine with considerable resources that can be virtualized.
IBM HMC Processor pool [cmdb_ci_processor_pool]	Virtual Machine Object [cmdb_ci_vm_object]	IBM shared pool used to allocate processors to a group of LPARs.

CMDB CI Class Models: Release 1.2.0 adds the following classes for the IBM HMC.

Class	Extends	Description
IBM HMC Server [cmdb_ci_hmc_server]	Server [cmdb_ci_server]	IBM console that manages frames and assigns logical partitions (LPARs) to pools.
IBM Frame [cmdb_ci_ibm_frame]	Server [cmdb_ci_server]	IBM physical machine with considerable resources that can be virtualized.
IBM LPAR Instance [cmdb_ci_lpar_instance]	Virtual Machine Instance [cmdb_ci_vm_instance]	IBM logical partition representing the virtual aspect of the operating system.
LPAR Resource [cmdb_ci_lpar_resource]	Configuration Item [cmdb_ci]	Resource of an LPAR instance.
IBM HMC Processor pool	Virtual Machine Object [cmdb_ci_vm_object]	IBM shared pool used to allocate processors to a group of LPARs.

Class	Extends	Description
[cmdb_ci_processor_pool]		

Class columns

CMDB CI Class Models: Release 1.3.0 adds the following columns to the respective classes.

IBM Frame [cmdb_ci_ibm_frame] class

Added columns	Description
Current available processor units	Current available processor units.
Configurable processor units	Configurable processor units.
Configurable memory units	Configurable memory units.
Installed processor units	Installed processor units.
Current available memory units	Current available memory units.
Installed memory units	Installed memory units.

IBM HMC Processor pool [cmdb_ci_processor_pool] class

Added columns	Description
LPAR IDs	LPAR IDs.
LPAR names	LPAR names.
Frame name	Frame name.

CMDB CI Class Models: Release 1.2.0 adds the following columns to the respective classes.

IBM HMC Server [cmdb_ci_hmc_server] class

Added columns	Description
Frame count	Count of frames.

Added columns	Description
Is VIO	Flags whether this VM is a (VIO) virtual input/output server in the HMC topology.
High Watermark VCPU	Peak in the utilization of virtual CPU assigned to during the reporting period.
Frame Serial Number	Serial number of a frame (frame being a physical machine such as ESX).
VIO Servers	List of VIO servers.

LPAR Resource [cmdb_ci_lpar_resource] class

Added columns	Description
Node name	Name of the node.
Partition Name	Name of the partition.
Partition Number	Number of the partition.
Type	Type of LPAR Resource.
Mode	Mode of LPAR Resource.
Entitled Capacity	Allotted capacity granted.
Partition Group-ID	ID of a group of partitions.
Shared Pool ID	ID of a pool of shared processors.

Added columns	Description
Online Virtual CPUs	A virtual CPU.
Maximum Memory	Maximum amount of memory.
Minimum Memory	Minimum amount of memory.
Variable Capacity Weight	Logical partition processor capacity weight.
Minimum Capacity	Minimum number of processes.
Capacity Increment	Increments of process.
Maximum Physical CPUs in system	Maximum CPUs allotted in system.
Active Physical CPUs in system	Current CPUs in system.
Active CPUs in Pool	Number of active CPUs within a pool.
Shared Physical CPUs in system	Number of shared CPUs within a system.
Maximum Capacity of Pool	Maximum capacity of processes within a pool.
Entitled Capacity of Pool	Number of processes that are entitled.
Unallocated Capacity	Number of free spaces.
Physical CPU Percentage	Number of CPUs allocated to system.
Unallocated Weight	At no extra charge resources on instance.
Desired Virtual CPUs	Target number of virtual CPUs.
Desired Memory	Target amount of memory.

Added columns	Description
Desired Variable Capacity Weight	Targeted processor load.
Desired Capacity	Target resources used within instance.
High Watermark VCPU	Peak in the utilization of virtual CPU assigned to during the reporting period.

IBM HMC Processor pool [cmdb_ci_processor_pool] class

Added columns	Description
Pool ID	ID of pool of processors.
CPU Core count	Number of CPU cores.
Memory count	Amount of memory used.
Unassigned cores	Number of unused cores.
Unassigned memory	Amount of unassigned memory.

The following class has no added columns: IBM Frame [cmdb_ci_ibm_frame].

Related concepts

- [CMDB schema model](#)

Internet of Things (IoT) extension classes

The CMDB CI Class Models store app adds or updates classes for the Internet of Things (IoT).

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as

Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

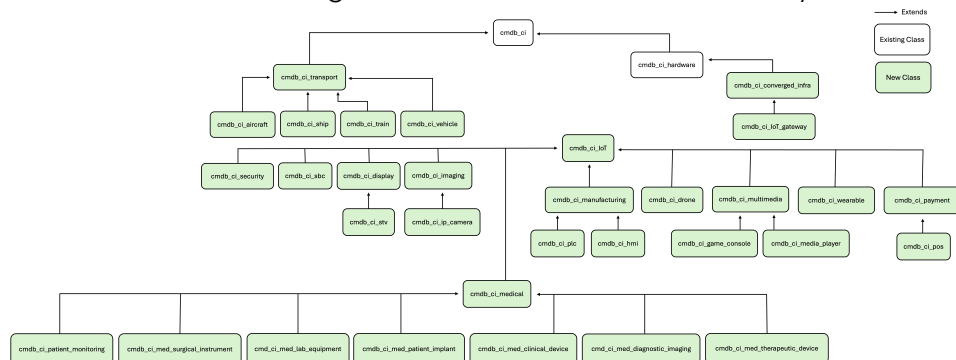
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Internet of Things (IoT)

IoT is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction. The classes added in this release, extend the Data Model to provide a foundation for the representation of IoT CI classes. This foundation underpins workflows for Enterprise Asset Management (EAM), Governance Risk Compliance (GRC), Component Supply Management (CSM), and Field Service Management (FSM) surrounding the management of IoT devices and the transport vehicles that some reside in.

IoT extension classes integrated with the CMDB class hierarchy



Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

CMDB CI Class Models: Release 1.6.0 adds the following classes for IoT.

Class	Extends	Description
Transport Type [cmdb_ci_transport]	cmdb_ci	Types of transportation that contain interconnected technology.
Converged Infrastructure [cmdb_ci_converged_infra]	cmdb_ci_hardware	Devices that serve both computing and networking functions.
IoT Device [cmdb_ci_iiT]	cmdb_ci_hardware	Parent table that contains Internet of Things device types.
Aircraft [cmdb_ci_aircraft]	Transport Type [cmdb_ci_transport]	A transportation method that utilizes air or space as its primary pathway. For example, airplanes and helicopters.
Ship [cmdb_ci_ship]	Transport Type [cmdb_ci_transport]	A transportation method that utilizes water as its primary pathway. For example, ships.
Train [cmdb_ci_train]	Transport Type [cmdb_ci_transport]	A transportation method that utilizes rails as its primary pathway. For example, Amtrak.
Vehicle	Transport Type [cmdb_ci_transport]	A transportation method that utilizes wheels or tracks as its

Class	Extends	Description
[cmdb_ci_vehicle]		method of movement. For example, cars, trucks, bulldozers.
IoT Gateway [cmdb_ci_IoT_gateway]	Converged Infrastructure [cmdb_ci_converged_infra]	A device that provides the following services: • Forwards packets between LAN and WAN on the IP layer. • Performs application layer functions between IoT nodes and other entities. • Enables local, short-range communication between IoT devices.
Security Device [cmdb_ci_security]	IoT Device [cmdb_ci_IoT]	Connected device that serves a security function such as badge readers.
Single Board Computing [cmdb_ci_sbc]	IoT Device [cmdb_ci_IoT]	Single Board Computing device such as a Raspberry Pi.
Display Device [cmdb_ci_display]	IoT Device [cmdb_ci_IoT]	Connected device that displays images.
Imaging Device	IoT Device [cmdb_ci_IoT]	Connected device that captures images.

Class	Extends	Description
[cmdb_ci_imaging]		
Medical Device [cmdb_ci_medical]	IoT Device [cmdb_ci_iot]	Connected device that serves a Medical Care function such as a nurse call unit.
Manufacturing Device [cmdb_ci_manufacturing]	IoT Device [cmdb_ci_iot]	Connected device that helps a manufacturing process.
Multimedia Device [cmdb_ci_multimedia]	IoT Device [cmdb_ci_iot]	A connected device that helps the generation or delivery of media content.
Payment Device [cmdb_ci_payment]	IoT Device [cmdb_ci_iot]	Connected device that allows for purchasing goods or services.
Drone [cmdb_ci_drone]	IoT Device [cmdb_ci_iot]	Unmanned connected device with mobility.
Wearable Technology [cmdb_ci_wearable]	IoT Device [cmdb_ci_iot]	Connected device that is worn by an entity such as a smart watch.
Smart Television [cmdb_ci_stv]	Display Device [cmdb_ci_display]	A television that is network connected and can run applications.
IP Camera	Imaging Device [cmdb_ci_imaging]	A camera that is network connected.

Class	Extends	Description
[cmdb_ci_ip_camera]		
Patient Monitoring [cmdb_ci_med_patient_monitoring]	Medical Device [cmdb_ci_medical]	Monitors and tracks various physiological parameters of a patient. Monitors heart rate, blood pressure, oxygen levels, respiratory rate, and so on.
Surgical Instrument [cmdb_ci_med_surgical_instrument]	Medical Device [cmdb_ci_medical]	Devices used during surgical interventions. Life support tools and tools to assist surgeons in precise and controlled maneuvers.
Lab Equipment [cmdb_ci_med_lab_equipment]	Medical Device [cmdb_ci_medical]	Specialized tools, apparatus, and instruments used in scientific research, experiments, and analyses.
Patient Implant [cmdb_ci_med_patient_implant]	Medical Device [cmdb_ci_medical]	Medical devices or tissues that are surgically placed inside a patient's body.
Clinical Device [cmdb_ci_med_clinical_device]	Medical Device [cmdb_ci_medical]	Devices used in a clinical setting for overall management, medical staff, and patient aid.
Diagnostic Imaging	Medical Device [cmdb_ci_medical]	Specialized equipment used in medicine to visualize

Class	Extends	Description
[cmd_ci_med_diagnostic_imaging]		the internal structures of the body.
Therapeutic Device [cmdb_ci_med_therapeutic_device]	Medical Device [cmdb_ci_medical]	Specialized tools designed to aid in the treatment, rehabilitation, or management of health conditions. Can range from orthopedic supports to cardiac implants and neurostimulation devices.
Process Logic Controller [cmdb_ci_plc]	Manufacturing Device [cmdb_ci_manufacturing]	A logic controller that is network connected which is used in manufacturing. For example, devices made by Siemens and Allen Bradley.
Human Machine Interface [cmdb_ci_hmi]	Manufacturing Device [cmdb_ci_manufacturing]	An HMI that is network connected which is used in manufacturing. For example, devices made by Siemens and Allen Bradley.
Game Console [cmdb_ci_game_console]	Multimedia Device [cmdb_ci_multimedia]	A device that is network connected which is used to play games or stream media. For example, an Xbox or Playstation.
Media Player	Multimedia Device [cmdb_ci_multimedia]	A device that is network connected

Class	Extends	Description
[cmdb_ci_media_player]		which is used to play digital media content. For example, Amazon Fire TV.
Display Monitor Control [cmdb_ci_monitor_control]	Multimedia Device [cmdb_ci_multimedia]	A device that is network connected that controls the display of media on a monitor. For example, a Crestron Media Controller.
Point of Sale Device [cmdb_ci_pos]	Payment Device [cmdb_ci_payment]	A device that is network connected which is used in the purchase of goods or services. For example, a credit card reader.

Related concepts

- [CMDB schema model](#)

Kong extension classes

The CMDB CI Class Models store app adds or updates classes for Kong gateways.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

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Kong

Kong is an API management platform that enables enterprise companies to better manage client and host traffic.

Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

CMDB CI Class Models: Release 1.49.0 adds the following classes for Kong.

Class	Extends	Description
Kong Gateway [cmdb_ci_kong_gateway]	API Gateway [cmdb_ci_api_gateway]	The Kong gateway application that hosts and manages individual APIs. Example: Kong Gateway instanceName.
Kong Load Balancer [cmdb_ci_kong_lb]	Load Balancer Application [cmdb_ci_lb_appl]	The default load balancer on the Kong gateway application that points to backend service instances when fulfilling API requests. Example: httpbin-upstream.
Kong Target	API Component	The load-balanced backend of the

Class	Extends	Description
[cmdb_ci_kong_target]	[cmdb_ci_api_component]	gateway that fulfills API requests. Example: httpbin-target1.

Class attributes

CMDB CI Class Models: Release 1.49.0 adds the following attributes to the respective classes.

Kong Gateway [cmdb_ci_kong_gateway]

Attribute	Data type	Description
Admin URL	String (255)	URL for making admin API requests.
Database	String	Type of database used by the Kong gateway. Example: Postgres or Cassandra.

Kong Load Balancer [cmdb_ci_lb_appl]

Attribute	Data type	Description
Algorithm	String	Type of algorithm used for load balancing. Example: round robin.
ID	String (255)	Unique identifier from the source system.

Kong Target [cmdb_ci_kong_target]

Attribute	Data type	Description
Target	String (255)	URL of target integration.

Key Relationship Structures

There are a number of key relationships that must be defined for API and Kong classes.

Kong relationships

Parent class	Relationship	Child class	Relationship type
API Backend [cmdb_ci_api_backend]	Uses::Used By	Kong Load Balancer	Suggested
Kong Load Balancer [cmdb_ci_lb_appl]	Contains::Contained By	Kong Target	Dependent
Kong Gateway [cmdb_ci_kong_gateway]	Provides::Provided By	Kong Load Balancer	Dependent

Related non-CMDB tables

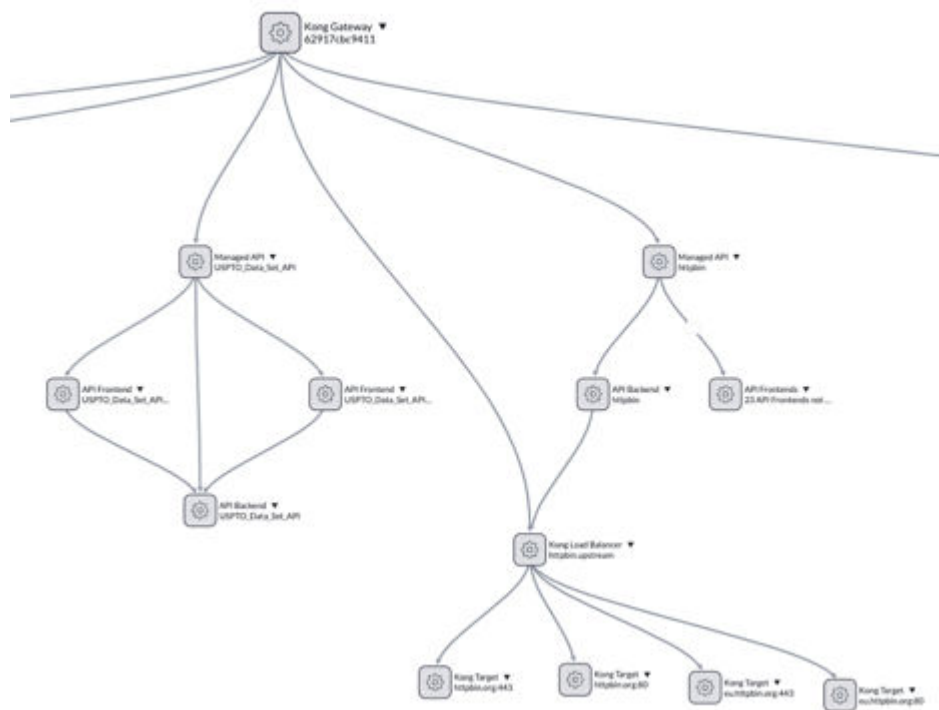
The Kong Gateway class uses the Kong Workspace non-CMDB table as a related list:

Kong Workspace [kong_workspace]

Attribute	Data type	Description
Name	String (100)	Name of the Kong workspace.
ID	String (255)	Unique identifier from the source system.
API Gateway	Reference	Reference to the Kong API gateway.

Kong gateway example

Here is an example of a dependency view for the Kong gateway class that shows how a gateway would populate the dependent managed API-dependent class with related APIs and components. The Managed API class is considered a first-level relationship with respect to the gateway, while the frontend and backend components are considered second-level relationships. From here, you can then bind alerts to these CIs, configure dynamic CIs for service views and incidents, or establish any additional workflows that use CIs.



Kubernetes extension classes

The Discovery and Service Mapping app adds or updates classes for the Kubernetes pattern.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

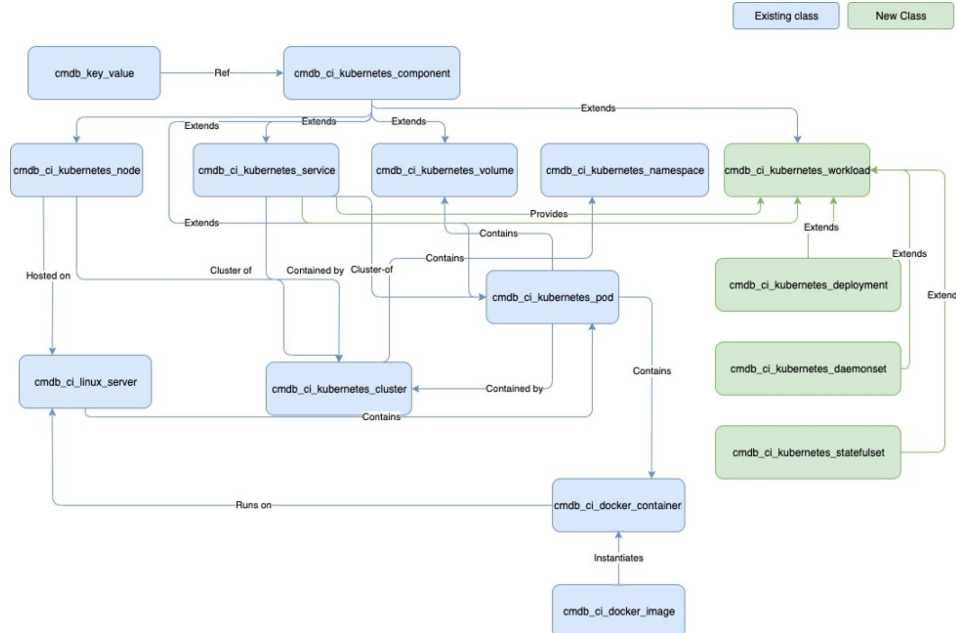
Request apps on the Store

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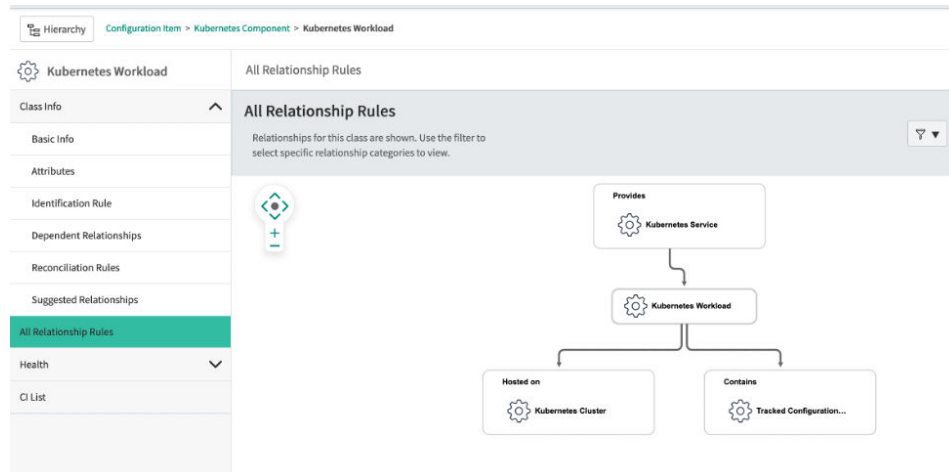
Kubernetes pattern

The Kubernetes pattern main flow helps with discovering Kubernetes core elements. The classes in this release, extend the support to discover Kubernetes workload controller components like deployments, daemonsets, and statefulsets. The Workload Share library captures information about deployments, daemonsets, and statefulsets and stores them in the respective tables. Other extensions include a YAML and service mesh extension that generates a YAML file to track configuration files and creating service to service relations by discovering service mesh information.

Kubernetes extension class integrated with the CMDB hierarchy



Kubernetes workload



Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates.

CMDB CI Class Models: Release 1.12.0 adds the following classes for Kubernetes pattern. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Extends	Fields	Relation
cmdb_ci_kubernetes_workload	cmdb_ci_kubernetes_components		Provides from cmdb_ci_kubernetes_service
cmdb_ci_kubernetes_deployment	cmdb_ci_kubernetes_workload	- Replicas Desired - Replicas Updated - Replicas Total - Replicas Available - Replicas Unavailable	Hosted on Cluster
cmdb_ci_kubernetes_daemonset	cmdb_ci_kubernetes_workload	- Pods running - Pods Waiting - Pods Succeeded - Pods Failed - Pods Available	Hosted on Cluster
cmdb_ci_kubernetes_statefulset	cmdb_ci_kubernetes_workload	- Pods running - Pods Waiting	Hosted on Cluster

Class	Extends	Fields	Relation
		- Pods Succeeded - Pods Failed - Pods Available	

Related concepts

- [CMDB schema model](#)

Network Intrusion Detection System (NIDS) CI extension class

The Network Intrusion Detection System (NIDS) [cmdb_ci_nids] class builds the relationships between passive network monitoring appliances, and the devices on the network that it discovers. A NIDS Manager manages the NIDS sensors that detect the devices and builds "detects::detected by" relationships between the NIDS records (parent) and the CIs it discovers (child).

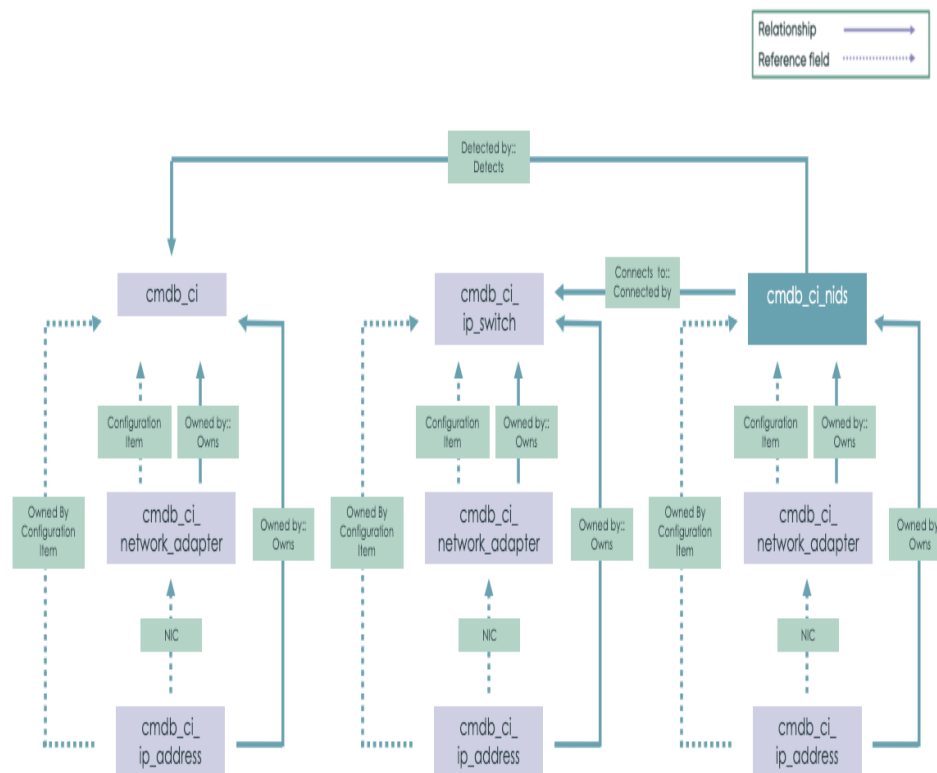
This topic lists the relevant classes that the CMDB CI Class Models store app adds or updates. See the class columns table for further details about the columns added for each class.

See the [full release notes for all CMDB CI Class Models](#).

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Network Intrusion Detection System (NIDS) schema structure



Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. See the class columns table for further details about the columns added for each class.

CMDB CI Class Models: Release 1.30 adds the following classes for the Network Intrusion Detection System (NIDS). For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Extends	Description
Network Intrusion Detection System (NIDS) (cmdb_ci_nids)	cmdb_ci_ids_network	NIDS is an intrusion detection system within the network that examines the traffic from all devices on the network. NIDS scanners build relationships between the OT network scanning appliances, and the OT devices on the network. An NIDS Manager manages the NIDS sensors.

Class columns

CMDB CI Class Models: Release 1.30 adds the following columns to the Network Intrusion Detection System (NIDS) [cmdb_ci_nids] class.

Network Intrusion Detection System (NIDS) [cmdb_ci_nids] class

Column label	Column name	Description
NIDS source ID	Correlation_id	Identifier of the NIDS device. Uses the assigned Correlation ID for the NIDS as its nids_source_id.
NIDS source name	nids_source_name	Name of the NIDS device.
Network type assignment	network_type_assignment	Designates if the device is on an IT or OT network.
NIDS assignment site	isa_entity_site	ISA site assigned to the NIDS. This information

Column label	Column name	Description
		is available when the logged in user has an assigned ISA Admin role.

Roles and Access Control Logic (ACLs)

The NIDS Admin (cmdb_nids_admin) role is associated with the Network Intrusion Detection System (NIDS) [cmdb_ci_nids] class: Can create, read, update, and delete Network IDS (NDIS) OT records. To view the Network IDS Application selection on the application menu, you must have this role.

Key relationship structure

For each CMDB CI record with a "Detected by" relationship with an NIDS record, a ServiceNow Operational Technology Certified Service Graph Connector does the following:

1. Assigns the following NIDS-related metadata values to the CI:
 - a. Location
 - b. Company
 - c. Related users (Owned by, Managed by, Supported by, Assigned to)
 - d. Related user groups (Approval group, Managed by Group, Support group, Change group)
2. If the NIDS network type is set to OT, it assigns the following NIDS-related metadata values to the CI:
 - a. Creates an OT device (cmdb_ot_entity) record for the CI, using the cmdb_ot_entity reference on the CI.
 - b. Assigns the NIDS assignment zone to the OT device record.
 - c. If the Industrial Process Manager is installed, assign the NIDS assignment site to the OT device record.

3. Life Cycle Stage and Life Cycle Stage Status values for the CI are used to capture the learning mode of a sensor.
 - If Life Cycle Stage is **Operational** and Life Cycle Stage Status is **Learning Mode**, then validation is unsuccessful.
 - If Life Cycle Stage Status is **In Use**, validation is successful.

Network Inventory (NI) extension classes

The CMDB CI Class Models store app adds or updates classes for the Telecommunications Network Inventory application. Telecommunications Network Inventory uses the Network Inventory (NI) extension classes to extend the Configuration Management Database (CMDB) Configuration Item (CI) class hierarchy.

These extensions enable the CMDB to store information about a service provider's network inventory. The store app adds class models that extend the CMDB class hierarchy, including:

- Class descriptions
- Identification rules
- Identifier entries
- Dependent relationships, if applicable.

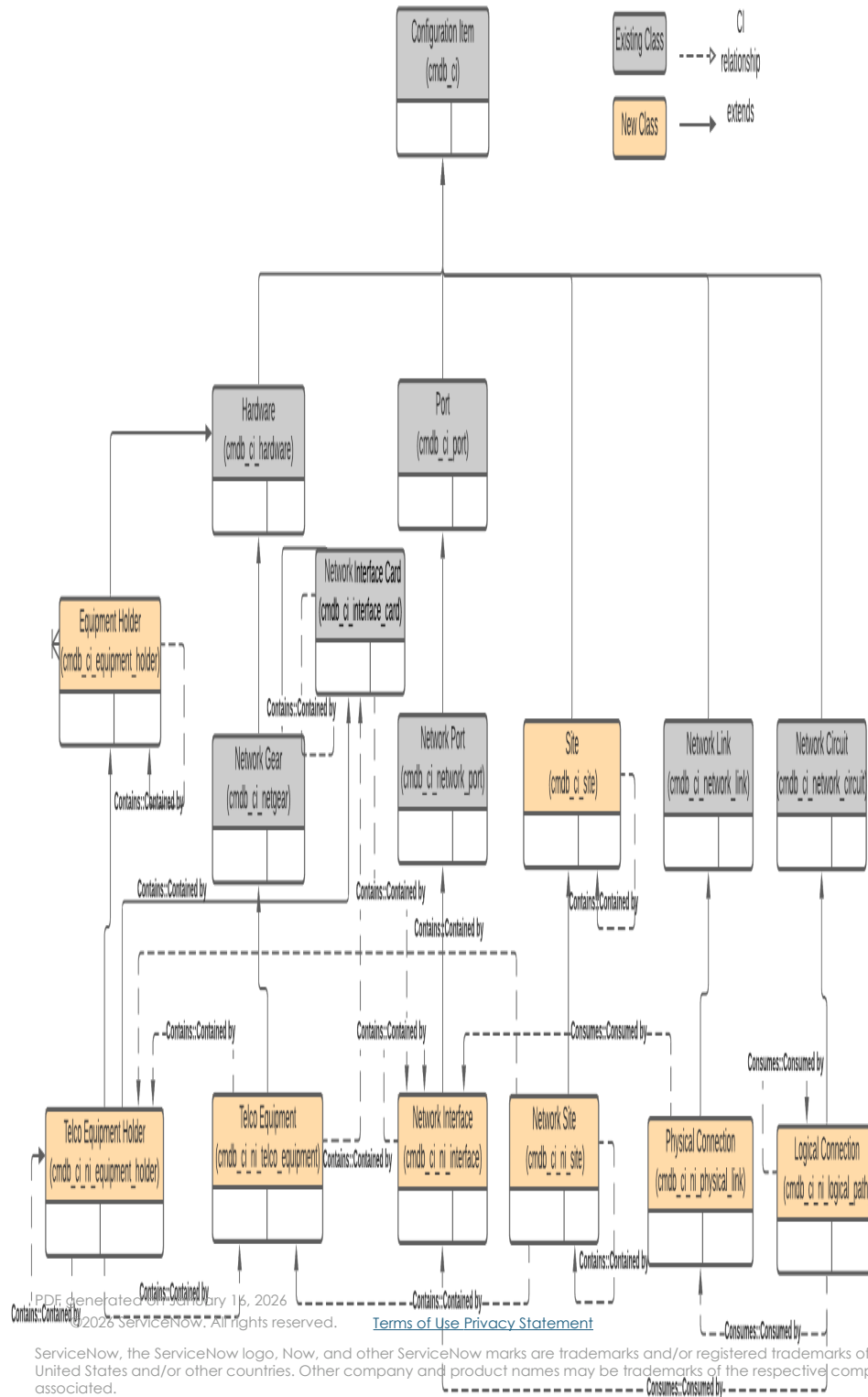
With the ServiceNow Telecommunications Network Inventory application, you can build a digital representation of your physical and logical networks. This application uses the NI class extensions to populate CIs that form the basis of your digital network inventory model. To learn more, see [Telecommunications Network Inventory](#).

See the [full release notes for all CMDB CI Class Models](#).

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Telecommunications Network Inventory (NI) schema structure



Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

NI extension classes

Class	Extends	Description
Network Site [cmdb_ci_ni_site]	Site [cmdb_ci_site]	Network Site. Captures and maintains the location-specific attributes for each network site, including the network centers, buildings, floors, and rooms where equipment is located. The network site records enable you to view all the equipment at a location. You can filter the locations by the assigned type, role, or function categories.
Telco Equipment Holder [cmdb_ci_ni_equipment_holder]	Equipment Holder [cmdb_ci_equipment_holder]	Telco Equipment Holder. Represents the physical units that contain the telecommunications equipment, including the cages, bays, cabinets, slots, relay racks, and line ups.

Class	Extends	Description
		The line ups contain the individual relay racks. Each relay rack contains the equipment shelves. The equipment holders can contain the other equipment holders. The equipment holder records enable you to track and manage your network assets.
Telco Equipment [cmdb_ci_ni_telco_equipmnet]	Network gear [cmdb_ci_netgear]	Telco Equipment. Represents a shelf or device that provides the technical functionality in a network. Examples include the routers, modems, mobile devices, optical cables, relays, and switches. The equipment can have slots, cards, or ports. The equipment can exist within an equipment holder or by itself because not all equipment is rack mounted. The equipment records enable you to track and manage your network assets.

Class	Extends	Description
Telco Filler Component [cmdb_ci_filler_component]	Hardware [cmdb_ci_hardware]	Filler Component. Represents a cover for empty spots/slots in a rack or device. Often made of flat metallic or plastic materials, helps separate cold and hot air zones, prevents dust from entering the chassis, and ensures proper airflow through the chassis or rack.
Telco Heat Baffle [cmdb_ci_heat_baffle]	Filler Component [cmdb_ci_filler_component]	Heat Baffle. Represents hardware used to help direct rising hot air away from equipment.
Telco Service Switching Point (SSP) [cmdb_ci_service_switching_point]	Telco Equipment [cmdb_ci_ni_telco_equipment]	Service Switching Point (SSP). Switch in a telecommunications network that sends a query to a central database called a service control point (SCP) via the SS7 network to determine how a TDM call can be routed. SSPs can be part of a voice switch or a separate

Class	Extends	Description
		computer connected to it.
Telco Service Control Point (SCP) [cmdb_ci_service_control_point]	Telco Equipment [cmdb_ci_ni_telco_equipment]	Service Control Point (SCP). Represents service control points (SCPs) in the Signaling System 7 (SS7) network responsible for routing calls and managing special features.
Telco Control Component [cmdb_ci_telco_control_component]	Telco Equipment [cmdb_ci_ni_telco_equipment]	Control Component. Device in a computer hardware component that creates an interface between a computer's main system motherboard and other hardware components. Some will be integrated directly into the motherboard, while others may be added on as expansion devices.
Network Interface [cmdb_ci_ni_interface]	Network Port [cmdb_ci_ni_network_port]	Network Interface. Captures and maintain equipment-

Class	Extends	Description
		specific attributes for the network interfaces. Identification: Serial Number (100) Name (200) Dependent on • Hardware (contains::contained by) • Logical Composite (contains::contained by) • Network Port (contains::contained by) • Virtual Machine Object (contains::contained by) • Kubernetes Component (contains::contained by) • Application Service (contains::contained by) Attribute of Network Port: MTU Size [mtu_size] In networking, maximum transmission unit (MTU) is a measurement

Class	Extends	Description
		representing the largest data packet that a network-connected device will accept. Measured in integer bytes. Use cases: - Optimizing network performance: Setting the right MTU size can reduce overhead and maximize data throughput. - Avoiding fragmentation: A properly configured MTU prevents packet fragmentation, reducing processing load and transmission delays. - Tuning for network type: Different networks (for example, VPNs, WANs) may require specific MTU sizes to ensure compatibility and optimal performance.
Network Interface Card (formerly Interface Card) [cmdb_ci_interface_card]	Hardware [cmdb_ci_hardware]	Network Interface Card. Represents interface cards that are stored

Class	Extends	Description
		in a network. Cards can occupy more than one slot and can contain other cards. They can be the equipment ports that are physical or logical (virtual). Each port is assigned a bandwidth value. The bandwidths are consumed when used in network designs.
Network Service Instance [cmdb_ci_network_service_instance]		Network Service Instance represents a specific deployed, provisioned, and/or configured instance of a set of network services that, in turn, are based on network functions.
Physical Connection [cmdb_ci_ni_physical_link]	Network Link [cmdb_ci_network_link]	Physical Connection. Represents the physical port connections on the network interface cards in your networks.
Logical Connection [cmdb_ci_ni_logical_path]	Network Circuit [cmdb_ci_network_circuit]	Logical Connection. Represents the logical or virtual port connections on the network interface cards. A logical connection typically

Class	Extends	Description
		represents the multiple physical connections on an interface card.

Nutanix extension classes

The CMDB CI Class Models store app adds or updates classes for Nutanix.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

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Nutanix Enterprise Cloud platform

The Nutanix Enterprise Cloud platform is a converged, scale-out compute and storage system that hosts and stores virtual machines. All nodes in a Nutanix cluster share the management of cluster resources. The foundational unit for the cluster is a Nutanix node which runs a standard hypervisor and contains processors, memory, and local storage (SSDs and hard disks). A Nutanix Controller virtual machine runs on each node, enabling the pooling of local storage from all nodes in the cluster.

Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. See the class columns table for further details about the columns added for each class.

CMDB CI Class Models: Release 1.6.0 adds the following classes for Nutanix. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Extends	Description
Nutanix Prism Central [cmdb_ci_nutanix_prism_central]	Virtual Machine Object [cmdb_ci_vm_object]	Multi-cluster manager responsible for managing multiple Acropolis Clusters to provide a single, centralized management interface.

CMDB CI Class Models: Release 1.2.0 adds the following classes for Nutanix.

Class	Extends	Description
Nutanix Cluster [cmdb_ci_nutanix_cluster]	Virtual Machine Object [cmdb_ci_vm_object]	Cluster comprising of the physical nodes running Nutanix software.
Nutanix Controller VM [cmdb_ci_nutanix_controller_vm]	Application [cmdb_ci_appl]	Nutanix controller virtual machine that is present in each node and that provides the storage clustering and management capabilities.
Nutanix Storage Container [cmdb_ci_nutanix_storage_container]	Storage Volume [cmdb_ci_storage_volume]	Subset of Nutanix storage pool used to apply policies such as reserved capacity, replication factor, and storage optimization options.

Class	Extends	Description
Nutanix Storage Pool [cmdb_ci_nutanix_storage_pool]	Storage Pool [cmdb_ci_storage_pool]	Grouping of physical disks within a Nutanix cluster which is typically used to create physical separation between virtual machines.
Nutanix Host [cmdb_ci_nutanix_host]	Virtualization Server [cmdb_ci_virtualization_server]	Physical host on which all the virtual machines run.
Nutanix Virtual Machine Instance [cmdb_ci_nutanix_vm_instance]	Virtual Machine Instance [cmdb_ci_vm_instance]	A virtual machine that runs on Nutanix infrastructure.

CMDB CI Class Models: Release 1.1.5 adds the following classes for Nutanix.

Class	Extends	Description
Nutanix Cluster [cmdb_ci_nutanix_cluster]	Virtual Machine Object [cmdb_ci_vm_object]	Cluster comprising of the physical nodes running Nutanix software.
Nutanix Controller VM [cmdb_ci_nutanix_controller_vm]	Application [cmdb_ci_app]	Nutanix controller virtual machine that is present in each node and that provides the storage clustering and management capabilities.

Class	Extends	Description
Nutanix Storage Container [cmdb_ci_nutanix_storage_container]	Storage Volume [cmdb_ci_storage_volume]	Subset of Nutanix storage pool used to apply policies such as reserved capacity, replication factor, and storage optimization options.
Nutanix Storage Pool [cmdb_ci_nutanix_storage_pool]	Storage Pool [cmdb_ci_storage_pool]	Grouping of physical disks within a Nutanix cluster which is typically used to create physical separation between virtual machines.
Nutanix Host [cmdb_ci_nutanix_host]	Virtualization Server [cmdb_ci_virtualization_server]	Physical host on which all the virtual machines run.
Nutanix Virtual Machine Instance [cmdb_ci_nutanix_vm_instance]	Virtual Machine Instance [cmdb_ci_vm_instance]	A virtual machine that runs on Nutanix infrastructure.

CMDB CI Class Models: Release 1.1.4 adds the following classes for Nutanix.

Nutanix Cluster [cmdb_ci_nutanix_cluster]	Virtual Machine Object [cmdb_ci_vm_object]	Cluster comprising of the physical nodes running Nutanix software.
Nutanix Controller VM	Application	Nutanix controller virtual machine that is

[cmdb_ci_nutanix_controller_vm]	[cmdb_ci_appl]	present in each node and that provides the storage clustering and management capabilities.
Nutanix Storage Container [cmdb_ci_nutanix_storage_container]	Storage Volume [cmdb_ci_storage_volume]	Subset of Nutanix storage pool used to apply policies such as reserved capacity, replication factor, and storage optimization options.
Nutanix Storage Pool [cmdb_ci_nutanix_storage_pool]	Storage Pool [cmdb_ci_storage_pool]	Grouping of physical disks within a Nutanix cluster which is typically used to create physical separation between virtual machines.
Nutanix Host [cmdb_ci_nutanix_host]	Virtualization Server [cmdb_ci_virtualization_server]	Physical host on which all the virtual machines run.
Nutanix Virtual Machine Instance [cmdb_ci_nutanix_vm_instance]	Virtual Machine Instance [cmdb_ci_vm_instance]	A virtual machine that runs on Nutanix infrastructure.

Class columns

CMDB CI Class Models: Release 1.6.0 adds no columns to the Nutanix Prism Central [cmdb_ci_nutanix_prism_central] class.

CMDB CI Class Models: Release 1.2.0 adds the following columns to the respective classes.

Nutanix Cluster [cmdb_ci_nutanix_cluster] class

Added columns	Description
Block Serial Numbers	Serial numbers of blocks that are connected to the cluster.
Cluster ID	UUID (Universal Unique Identifier) of the cluster.
External Subnet	Subnet of the external IP address of the cluster.
Full Version	Full version of the cluster. For example: el7.3-release-euphrates-5.10.3.1-stable-655d4def34bf18785782f2adb8cdd5f8457d1fe3
Hypervisor Types	Types of hypervisors that are related to this cluster.
Internal Subnet	Subnet of internal IP addresses.
NCC Version	Nutanix cluster check version.
NTP Servers	NTP servers that are related to this cluster.
Number of Nodes	Number of nodes that are connected to the cluster.
Timezone	Timezone of the cluster.
Version	Version of the cluster. For example: 5.10.3.1

Nutanix Controller VM [cmdb_ci_nutanix_controller_vm] class

Added columns	Description
Hypervisor Type	Type of hypervisor.
Memory (MB)	Amount of memory (in MB) available on the controller.
State	On/off power state of controller.
VM ID	UUID of the controller virtual machine.
Object ID	ID of the controller virtual machine.

Nutanix Storage Container [cmdb_ci_nutanix_storage_container] class

Added columns	Description
Compression	Indicates whether compression is enabled.
Container ID	UUID of the container.
Deduplication	Indicates whether on disk deduplication is enabled, that is dedup compression applied to data on hard disks (HDD). Performance tier deduplication is a prerequisite for on disk deduplication.
Erasur Code	Indicates whether erasure coding is enabled.
Replication Factor	Number of maintained data copies. The replication factor is

Added columns	Description
	specified (2 or 3) when the container is created.

The following classes have no added columns:

- Nutanix Storage Pool [cmdb_ci_nutanix_storage_pool]
- Nutanix Host [cmdb_ci_nutanix_host]
- Nutanix Virtual Machine Instance [cmdb_ci_nutanix_vm_instance]

CMDB CI Class Models: Release 1.1.5 adds the following columns to the respective classes.

Nutanix Cluster [cmdb_ci_nutanix_cluster] class

Added columns	Description
Block Serial Numbers	Serial numbers of blocks that are connected to the cluster.
Cluster ID	UUID (Universal Unique Identifier) of the cluster.
External Subnet	Subnet of the external IP address of the cluster.
Full Version	Full version of the cluster. For example: el7.3-release-euphrates-5.10.3.1-stable-655d4def34bf18785782f2adb8cdd5f8457d1fe3
Hypervisor Types	Types of hypervisors that are related to this cluster.
Internal Subnet	Subnet of internal IP addresses.
NCC Version	Nutanix cluster check version.

Added columns	Description
NTP Servers	NTP servers that are related to this cluster.
Number of Nodes	Number of nodes that are connected to the cluster.
Timezone	Timezone of the cluster.
Version	Version of the cluster. For example: 5.10.3.1

Nutanix Controller VM [cmdb_ci_nutanix_controller_vm] class

Added columns	Description
Hypervisor Type	Type of hypervisor.
Memory	Amount of memory (in MB) available to the virtual machine.
State	On/off state of power.
VM ID	UUID of the controller virtual machine.
Object ID	ID of the controller virtual machine.

Nutanix Storage Container [cmdb_ci_nutanix_storage_container] class

Added columns	Description
Compression	Indicates whether compression is enabled.
Container ID	UUID of the container.

Added columns	Description
Deduplication	Indicates whether on disk deduplication is enabled, that is dedup compression applied to data on hard disks (HDD). Performance tier deduplication is a prerequisite for on disk deduplication.
Erasure Code	Indicates whether erasure coding is enabled.
Replication Factor	Number of maintained data copies. The replication factor is specified (2 or 3) when the container is created.

The following classes have no added columns:

- Nutanix Storage Pool [cmdb_ci_nutanix_storage_pool]
- Nutanix Host [cmdb_ci_nutanix_host]
- Nutanix Virtual Machine Instance [cmdb_ci_nutanix_vm_instance]

CMDB CI Class Models: Release 1.1.4 adds the following columns to the respective classes.

Nutanix Cluster [cmdb_ci_nutanix_cluster] class

Added columns	Description
Block Serial Numbers	Serial numbers of blocks that are connected to the cluster.
Cluster ID	UUID (Universal Unique Identifier) of the cluster.

Added columns	Description
External Subnet	Subnet of the external IP address of the cluster.
Full Version	Full version of the cluster. For example: el7.3-release-euphrates-5.10.3.1-stable-655d4def34bf18785782f2adb8cdd5f8457d1fe3
Hypervisor Types	Types of hypervisors that are related to this cluster.
Internal Subnet	Subnet of internal IP addresses.
NCC Version	Nutanix cluster check version.
NTP Servers	NTP servers that are related to this cluster.
Number of Nodes	Number of nodes that are connected to the cluster.
Timezone	Timezone of the cluster.
Version	Version of the cluster. For example: 5.10.3.1

Nutanix Controller VM [cmdb_ci_nutanix_controller_vm] class

Added columns	Description
Hypervisor Type	Type of hypervisor.
Memory	Amount of memory (in MB) available to the virtual machine.

Added columns	Description
State	On/off state of power.
VM ID	UUID of the controller virtual machine.
Object ID	ID of the controller virtual machine.

Nutanix Storage Container [cmdb_ci_nutanix_storage_container] class

Added columns	Description
Compression	Indicates whether compression is enabled.
Container ID	UUID of the container.
Deduplication	Indicates whether on disk deduplication is enabled, that is dedup compression applied to data on hard disks (HDD). Performance tier deduplication is a prerequisite for on disk deduplication.
Erasure Code	Indicates whether erasure coding is enabled.
Replication Factor	Number of maintained data copies. The replication factor is specified (2 or 3) when the container is created.

The following classes have no added columns:

- Nutanix Storage Pool [cmdb_ci_nutanix_storage_pool]
- Nutanix Host [cmdb_ci_nutanix_host]

- Nutanix Virtual Machine Instance [cmdb_ci_nutanix_vm_instance]

Related concepts

- [CMDB schema model](#)

OpenStack extension classes

The CMDB CI Class Models store app adds or updates classes for OpenStack.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

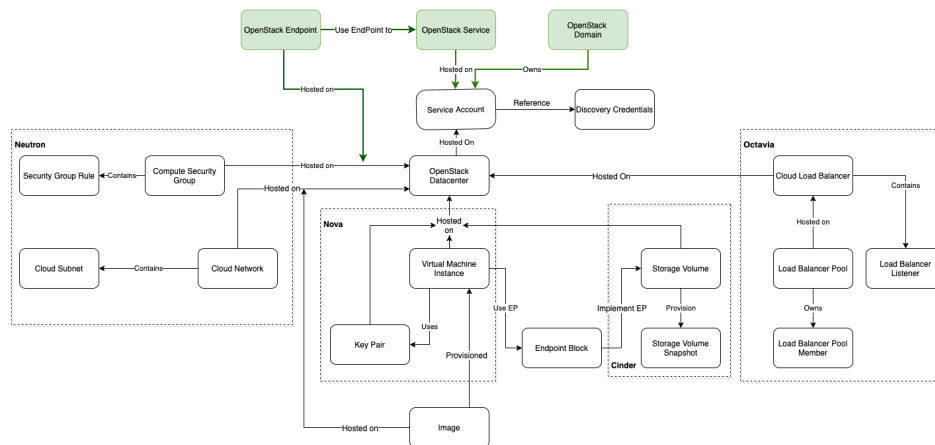
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OpenStack

OpenStack is a cloud operating system that controls large pools of compute, storage, and networking resources throughout a datacenter. All of these resources are managed and provisioned through APIs with common authentication mechanisms. Other components provide services such as orchestration, fault management, and service management to ensure high availability of user applications. OpenStack is broken up into services to enable you to plug and play components depending on your needs. These components are designed for horizontal scalability, so you can easily add new resources to grow your cloud over time.

OpenStack classes integrated with the CMDB class hierarchy



Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. See the class columns table for further details about the columns added for each class.

CMDB CI Class Models: Release 1.8.0 adds the following classes for OpenStack. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Extends	Description
OpenStack Services [cmdb_ci_cloud_open_stack_service]	Virtual Machine Object [cmdb_ci_vm_object]	An OpenStack web service that can be accessed via a URL.
OpenStack Endpoint [cmdb_ci_cloud_open_stack_endpoint]	Virtual Machine Object [cmdb_ci_vm_object]	The access point of a Service.
OpenStack Domain [cmdb_ci_cloud_open_stack_domain]	Virtual Machine Object [cmdb_ci_vm_object]	A collection of users, groups, and projects.

Class columns

CMDB CI Class Models: Release 1.8.0 adds the following columns to the respective classes.

OpenStack Services [cmdb_ci_cloud_openstack_service] class

Added columns	Description
type	The Service type, which describes the API implemented by the Service. Possible values: Compute, ec2, identity, image, network, or volume.
enabled	Defines if the service and its endpoints appear in the Service catalog (true/false).

OpenStack Endpoint [cmdb_ci_cloud_openstack_endpoint] class

Added columns	Description
interface	The interface type, which describes the visibility of the endpoint. Possible values: • public - Visible by end users on a publicly available network interface. • internal - Visible by end users on an unmetered internal network interface. • admin - Visible by administrative users on a secure network interface.
enabled	Defines if the Service and its endpoints appear in the Service catalog (true/false).

OpenStack Domain [cmdb_ci_cloud_openstack_domain] class

Added columns	Description
enabled	Defines if the domain is enabled (true/false).

Related concepts

- [CMDB schema model](#)

Operational Technology (OT) extension classes

The CMDB CI Class Models store app adds or updates classes for Operational Technology (OT).

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

Operational Technology (OT) data model

The Operational Technology (OT) data model was created to enable management of “OT Device” data. Operational Technology that controls industrial equipment can be based on IT class hardware (computers, servers, network gear, and so on), or on specific hardware profiles not included in the ServiceNow IT class model (PLCs, HMIs, Engineering Workstations, Historians, and so on). Therefore, a single OT Device in the OT data model includes two primary components:

1. A CI class record. This can be an IT or an OT class CI.
2. An OT Device details record. This describes the OT Device type (function) and other OT-specific attributes.

Each OT Device in the CMDB can be distinguished as having an “OT Device Details” reference [cmdb_ot_entity reference field] to a

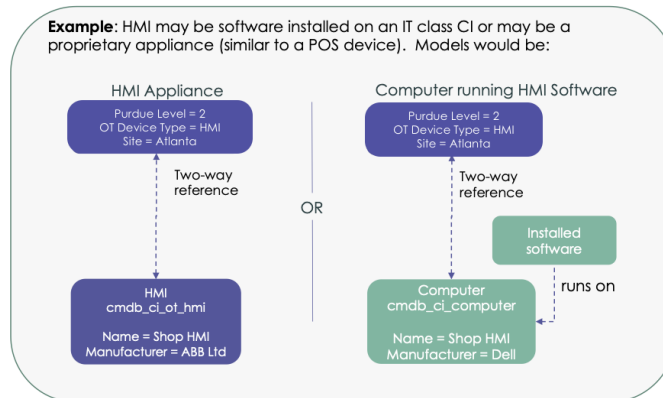
specific OT Device details [cmdb_ot_entity table] record. This is a bi-directional reference; the ot_device reference on the cmdb_ot_entity table references the CI record. If the cmdb_ot_entity reference of a given CI record is **not** empty, the CI is considered to be an OT Device.

For example, an HMI (Human Machine Interface) OT Device could be composed in at least two different ways.

1. A computer CI with an OT Device detail record describing its "OT Device type" as HMI.
2. An HMI CI with an OT Device detail record describing its "OT Device type" also as HMI.

A record in any CI class can be designated as an "OT Device"

All OT Device records include a record in the OT Entity table (cmdb_ot_entity) to store OT-specific metadata.



In this way, a list of HMIs can be derived and a list of all computers in an enterprise can also be derived.

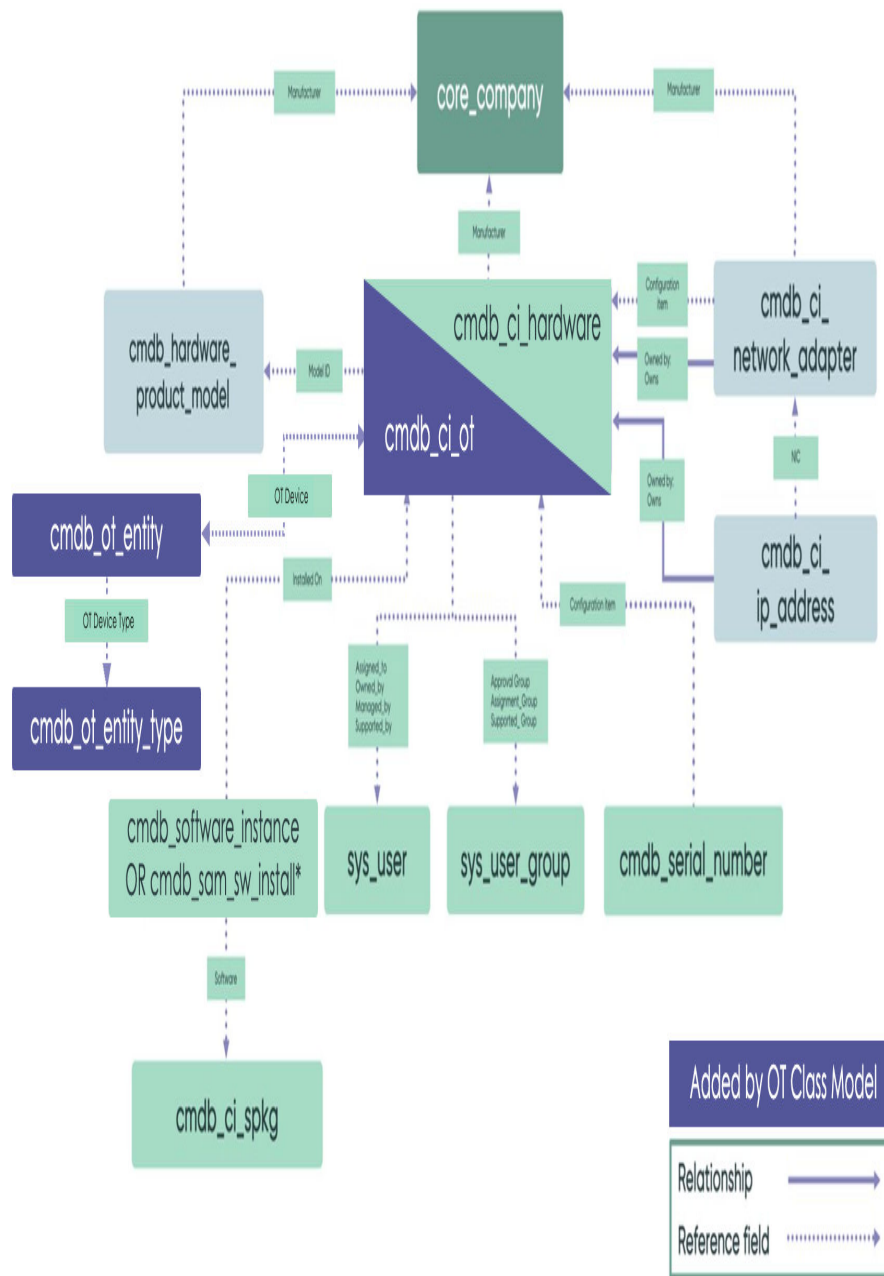
For more details on the OT data model, see [Implementing the CSDM framework for Operational Technology](#).

You can use the added classes as any other CMDB class. Applications such as Discovery for Operational Technology, and Service Graph Connector for Operational Technology (Excel) use these class extensions to populate CIs and discover various technologies and software. To learn more, see:

- Service Graph Connector for Microsoft Excel
- IT Discovery for OT Networks

Note: In Operational Technology, CIs used on an OT network to automate an industrial process are often referred to as OT Devices. This term shouldn't be confused with a Device record commonly used in the practice of Asset Management.

Operational Technology (OT) schema structure



* if software asset management is installed

Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. See the class columns table for further details about the columns added for each class. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

CMDB CI Class Models: Release 1.69.0 adds the following classes for Operational Technology (OT).

Class	Description
Firmware Installation [cmdb_firmware_install]	Firmware associated with the OT device.
Key Value [cmdb_key_value]	Additional information related to the OT device that's populated with the available OT integrations and captured as Key value pairs.
Software Instance [cmdb_sam_sw_install]	Captures the software installed on the OT device if Software Asset Management isn't available. Note: If Software Asset Management is installed and entitled, the software installed data is available in the Software Installation related list.

CMDB CI Class Models: Release 1.59.0 adds the following classes for OT.

Class	Extends	Description
OT System Service [cmdb_ci_ot_system_service]	cmdb_ci_ot_system	Category of technology and systems that are used to manage, control, and monitor physical processes,

Class	Extends	Description
		machinery, and industrial operations.

CMDB CI Class Models: Release 1.56.0 adds the following classes for OT.

Class	Extends	Description
Quality Inspection Control System [cmdb_ci_ot_qics]	cmdb_ci_ot_control	Control systems that assist specifically in quality and inspection functions.

CMDB CI Class Models: Release 1.30 adds the following classes for OT.

Class	Extends	Description
CNC [cmdb_ci_ot_cnc]	cmdb_ci_ot_control	Computer Numerical Control, used for automated control of machining tools such as drills, lathes, mills, and for 3D printers.
DCS [cmdb_ci_ot_dcs]	cmdb_ci_ot_control	Distributed Control System. Achieves control using intelligence distributed about the controlled process, rather than by a centrally located single unit.
DPU [cmdb_ci_ot_dpu]	cmdb_ci_ot_control	Distributed Processing Units. ICS on a dedicated network, with each DPU handling thousands of points of I/O.
EWS [cmdb_ci_ot_ews]	cmdb_ci_ot_supervisory	Engineering Workstation. A computing platform

Class	Extends	Description
		for configuration, maintenance, and diagnostics of ICS applications and other control system equipment.
Historian [cmdb_ci_ot_historian]	cmdb_ci_ot_supervisory	Data Historian. A centralized database supporting data analysis for industrial processes.
HMI [cmdb_ci_ot_hmi]	cmdb_ci_ot_supervisory	Human-Machine Interface. Hardware or software through which an operator interacts with a controller.
IED [cmdb_ci_ot_ied]	cmdb_ci_ot_control	Intelligent Electronic Device. Receives or sends data/control from or to an external source for power grids.
Industrial Actuator [cmdb_ci_ot_industrial_actuator]	cmdb_ci_ot_field_device	Component of a machine that is responsible for moving and controlling a mechanism, such as opening a valve.
Industrial Drive [cmdb_ci_ot_industrial_drive]	cmdb_ci_ot_field_device	Equipment used to control the speed of machinery. It can be a mechanical electromechanical, hydraulic, or electronic device.

Class	Extends	Description
Industrial 3D Printer [cmdb_ci_ot_industrial_3d_printer]	cmdb_ci_ot_control	Device used in additive manufacturing for the construction of a three-dimensional object from a CAD model, or a digital 3D model.
Industrial Robot [cmdb_ci_ot_industrial_robot]	cmdb_ci_ot_field_device	Robotic system used for manufacturing.
Industrial Sensor [cmdb_ci_ot_industrial_sensor]	cmdb_ci_ot_field_device	Sensor device used to monitor the health of equipment
Network Gear [cmdb_ci_netgear]	cmdb_ci_hardware	Network gear is an electronic device which is required for communication and interaction between devices on a computer network.
Operations Technology [cmdb_ci_ot]	cmdb_ci_hardware	Base class for Operational Technology, used for industrial control. For instance, in manufacturing.
OPC Client [cmdb_ci_ot_opc_client]	cmdb_ci_ot_supervisory	Software module that enables applications to acquire data from an OPC Server or conduct supervisory control using an OPC Server.

Class	Extends	Description
OPC Server [cmdb_ci_ot_opc_server]	cmdb_ci_ot_control	Software module that enables applications to provide their data to the outside world using OPC.
OT Control Module [cmdb_ci_ot_control_module]	cmdb_ci_ot_control	Module such as a PLC or DCS connected to an OT Control System.
OT Control System [cmdb_ci_ot_control]	cmdb_ci_ot	Base Class for industrial control systems (ICS), usually at Purdue Model Level 1 or 2.
OT Supervisory System [cmdb_ci_ot_supervisory]	cmdb_ci_ot	Base class for supervisory systems, usually at Purdue Model Level 2 or 3.
PLC [cmdb_ci_ot_plc]	cmdb_ci_ot_control	Programmable Logic Controller. Used to control OT Devices.
Protocol Converter [cmdb_ci_protocol_converter]	cmdb_ci_hardware_network_gear	Device used to convert standard or proprietary protocol of one device to the protocol suitable for the other device or tools to achieve the interoperability.
RTU [cmdb_ci_ot_rtu]	cmdb_ci_ot_control	Remote Terminal Unit. Special purpose data acquisition and control unit designed to support DCS and SCADA remote stations

Class	Extends	Description
SCADA Client [cmdb_ci_ot_scada_client]	cmdb_ci_ot_supervisory	Supervisory Control and Data Acquisition. Client that enables an operator to manage a SCADA server.
SCADA Server [cmdb_ci_ot_scada_server]	cmdb_ci_ot_control	Supervisory Control and Data Acquisition. System capable of gathering and processing data and applying operational controls over long distance.

Class columns

CMDB CI Class Models: Release 1.30 adds the following columns to the respective classes.

Operational Technology (OT) [cmdb_ci_ot] class

Added columns	Description
firmware_version	Firmware version reported by discovery source.
hardware_version	Hardware version reported by discovery source.

OT Control System [cmdb_ci_ot_control_system] class

Added columns	Description
has_module	true/false value describing if the system has modules, such as chassis / blade architecture.
backplane_name	System-reported string name or number for the backplane.

Added columns	Description
backplane_id	System-reported unique ID for the backplane

OT Control Module [cmdb_ci_ot_control_module] class

Added columns	Description
slot_number	Reported slot in the control system this module is using.
module_type	Module type reported by the discovery source.

OT Field Device [cmdb_ci_ot_field_device] class

Added columns	Description
device_type	List that describes if the device provides input, output, or both to the parent control system.

OT System Service [cmdb_ci_ot_system_service] class

Added columns	Description
display_name	Name of the OT system.
isa_entity_site	Site where the OT system is located.

Form view

All Operational Technology (OT) extension classes have a "Default view" form view that includes the OT Device Details attribute at the top of the form. The following table lists the other classes that have the **Operational Technology (OT)** view on their form context menu. This is the default form view for users with the cmdb_ot_viewer role for the following classes.

Class	Description
base hardware [cmdb_ci_hardware]	Base class for hardware.
base computer [cmdb_ci_computer]	An extension of the Hardware table, capturing computer properties.
base server [cmdb_ci_server]	Base class for all types of servers.
linux server [cmdb_ci_linux_server]	Server running Linux software.
windows server [cmdb_ci_win_server]	A server running Microsoft Windows Server operating system.
IoT Device base class [cmdb_ci_iot]	Parent table that contains Internet of Things device types.
IP Firewall and extended classes [cmdb_ci_ip_firewall]	Contains all network firewalls.
IP Router [cmdb_ci_ip_router]	Specialization of the Network Gear [cmdb_ci_netgear] table.
IP Switch [cmdb_ci_ip_switch]	Specialization of the Network Gear [cmdb_ci_netgear] table.
Protocol Converter [cmdb_ci_protocol_converter]	Device used to convert standard or proprietary protocol of one device to the protocol suitable for the other device or tools to achieve the interoperability.

Classes not included in the table do not have the Operational Technology (OT) view by default. For any additional classes required, you can add the Operational Technology (OT) view to the form context menu. For more information about form context menu options, see [Form context menu](#).

Roles and Access Control Logic (ACLs)

The following roles are associated with the Operational Technology (OT) [cmdb_ci_ot] classes and associated tables that follow:

Admin (cmdb_ot_admin)

Can create, read, update, and delete OT records. An admin can also edit the OT Device Type on the OT Device form, and manage specific configurations on the OT entity tables.

Editor (cmdb_ot_editor)

Can create, read, update, and delete OT Device records.

Viewer (cmdb_ot_viewer)

Can read OT Device records.

Note: For the cmdb_ci_ot and cmdb_ot_entity tables:

- A user must have one of these three OT roles to view OT Devices.
- IT users with an assigned itil role are restricted from viewing OT Devices in the cmdb_ci_ot table, and records in the cmdb_ot_entity table. IT users are still able to see IT asset classes, such as Computer, Installed Software.

OT customers may want to restrict access to OT Devices from users with an IT (itil) only role, for both a viewer and an admin role.

Key relationship structures

Use the following key relationships as important guidelines when creating Operational Technology (OT) CIs:

OT Entity

Since any CI class may be found on an OT network, the OT Entity table [cmdb_ot_entity] captures additional attributes required in an OT Environment:

Attribute	Description
Business criticality	Business criticality assigned in the discovery source.
OT Device	Reference to the CI that is on the OT network.
OT Device type	The function of the OT Device, regardless of CI class. - For a dedicated HMI appliance, the CI is in the <code>cmdb_ci_ot_hmi</code> class, and the OT Device type should be HMI. - For a computer performing the function of an HMI, the CI is in the <code>cmdb_ci_computer</code> class and the OT Device type should be HMI
Purdue level	Purdue level of the OT Device.
Zone	Zone assigned to the OT Device, usually used in the context of a zone/conduit model.
IRE criterion attribute (<code>ire_criterion</code>)	Acts as a criterion attribute for an OT entity-related entry.

The OT Entity is a related list added to the Operational Technology (OT) [`cmdb_ci_ot`] table and extended tables. If you want to view OT entity metadata on an existing CI class, first add the related list to the form.

OT Entity Type

The OT Entity Type [`cmdb_ot_entity_type`] table tracks the type of OT Device that an OT or non-OT CI is performing the function of. It serves as a necessary part of the Purdue level data model, and extends the Application File [`sys_metadata`] table.

The ServiceNow AI Platform includes records representing common OT Device types of OT Device Cls. The OT Entity Type table attributes include:

Attribute	Description
Label	Display name of a specific OT Device type.
Name	Value used to identify a specific OT Device type.
OT table	Value used for auto-populating the Type field for the OT entity record [cmdb_ot_entity] of an OT Class Cl.
Parent	Parent type of a specific type, which is a reference to a record in the same table [cmdb_ot_entity_type].

OT Class Mapping Template

The OT Class Mapping Template [ot_class_mapping_template] table extends the Application File [sys_metadata] and is used by OT Service Graph Connectors (SGCs) to classify a device into the correct CI class based on the source data.

Each SGC can extend the OT Class Mapping Template table and create a child class in the SGC application. Doing so defines the SGC-specific device classification details. For example, the Excel SGC Class Mapping [excel_sgc_class_mapping] table is a child of the OT Class Mapping Template created in the Service Graph Connector for Microsoft Excel application scope, which stores the classification details specific for the Service Graph Connector for Microsoft Excel.

Note: The OT Class Mapping Template doesn't store any classification detail records. The child tables specific to each SGC store the classification records for an SGC to use.

The OT Class Mapping Template table attributes include:

Attribute	Description
Source Class	The class of the source CI.
Target CMDB class	The expected ServiceNow class that for the CI.
OT Device type	The category type that the OT Device is classified as. The Device type is also the function that the Device plays on the OT network. For example: An IT Device, such as a server, can be converted to an OT Device, and the function it plays on the network is an HMI. Therefore, its class is server and its Device type is HMI. Note: In some cases, there are OT Devices with no OT function or OT Devices where the Device type is unknown. For OT Devices with no OT function, select No OT Function. For OT Devices where the Device type is unknown, select Unknown.
Allow OS classification	When set to True, if an operating system is found on the CI, the target is switched away from this table's target class to a ServiceNow class that matches its OS. For example, if Historian is the target class but Windows Server OS is found in the payload, the CI is classified as a Windows Server CI

Attribute	Description
	and its OT device type is set to Historian.
Active	When checked, the class mapping is set to Active and considered during device classification.

Serial number

During CI identification, the Identification and Reconciliation Engine (IRE) processes search for a serial number in two locations. One is the CI serial number attribute, and the other is the Serial Number [cmdb_serial_number] table, with reference back to the Operational Technology (OT) [cmdb_ci_ot] table.

- It stores any serial number of any type other than system serial number, and only in the Serial Number table (and not in the server CI attribute).
- If the system serial number is available, it stores it in both the Serial Number attribute of the CI and in the Serial Number table.
- The Serial Number table is a many-to-one relationship, linking back to the Operational Technology (OT) CI. This table has a Type field for specifying the type of the serial number.
- For Operational Technology (OT), use the string value system for the serial number type to ensure proper reconciliation across various sources.

Network adapter

Use the Network Adapter [cmdb_ci_network_adapter] class to store network adapters.

1. Set the MAC Address attribute to be the MAC address value.
 - Format the string with colon separators between octets and lower case hexadecimal characters with padded zeros.
 - For example, 'f8:f2:1e:00:d4:66'.
2. Set the Name attribute in the Network Adapter class to be the same as the MAC Address.

3. In the CI Relationship [cmdb_rel_ci] table, create an Owned By:Owns relationship to the associated Hardware CI.
4. Using the CI with a reference to the associated Hardware CI, specify a reference from the Network Adapter [cmdb_ci_network_adapter] table.

IP Address

Use the IP Address [cmdb_ci_ip_address] class to store IP addresses.

1. Store an IP address value in the IP Address attribute, and in the Name attribute to avoid empty Name attributes.
 - Store an IPv4 IP address value using the format 'NNN.NNN.NNN.NNN', with decimal-based octets and period separators. Non-conforming values should be considered invalid and cleansed to null values.
 - Store an IPv6 IP address value using lower case hexadecimal with colon separators. Non-confirming values should be considered invalid and cleansed to null values.
2. Set the Netmask attribute to the IP address.
3. Set the Name attribute in the Network Adapter class to be the same as the MAC Address.
4. In the CI Relationship [cmdb_rel_ci] table, create an Owned By::Owns relationship to the associated Hardware CI.
5. In the CI Relationship [cmdb_rel_ci] table, create an Owned By::Owns relationship to the associated Hardware CI.
6. For the IP address, specify a reference to the Network Adapter [cmdb_ci_network_adapter] table, using the Configuration Item with a reference to the associated Hardware CI.
7. To ensure that base system identification rules work properly, also store the IP address in the associated Network Adapter class.

Network adapter and IP address

Use the IP Address [cmdb_ci_ip_address] class to store IP addresses.

1. Store the MAC address of the network adapter installed on a server, in the Network Adapter [cmdb_ci_network_adapter] class.

2. Store the IP address in the IP Address [cmdb_ci_ip_address] class.

Note: Do not store the MAC address or the IP address in the Operational Technology (OT) [cmdb_ci_ot] table. The default Operational Technology (OT) form is configured to display the IP address from the Network Adapter table.

Backup Storage Information

Use the Backup Storage Information [cmdb_backup_storage_information] class to create backup records that identify the device in internal or external backup services. The Backup Storage Information table includes the following attributes:

Attributes	Description
Backup Configuration Status	Indicates whether the CI has been configured in the backup service or appliance with relevant policies.
Discovery Source	Discovery source of the backup record.
Backup Service	Reference to the Common Service Data Model Technical Service hosting external and internal backup system, services, or appliances
Managed By	Email ID of the user responsible for managing the backup.
Latest Successful Execution	Date and time of the most recent data backup that was successfully executed for a device.
Latest Execution Attempt	Date and time of the most recent data backup that was attempted for a device.
Next Execution Attempt	Date and time when the next data backup is going to be attempted for a device.

Attributes	Description
Latest Execution Log	Detailed log entries for all device data backup that were executed.
Backup Source ID	Backup service record for a device, which identifies the device in external or internal backup services. Backup source id can include host_id, vcenter_id, instance_id, db_id.
Job Definition ID	Identification number assigned to a data backup job.
Recovery Point Objective	Amount of time that can elapse between backups and the amount of data lost.
Execution Mode	Indicates whether the backup is configured to run automatically on a periodic basis, or if it is manually executed on an as-needed basis.
Backup Source	Details of the backed up CI. This table holds the backup status details for the referenced CI.
Job Definition Name	Name assigned to a data backup job.
Managed By Group	Primary support group responsible for managing the backup.
Domain	ID of the domain to which the instance belongs.

Use the Backup Job Execution History [cmdb_backup_job_execution_history] class to create a history of records for all backup executions. The Backup Job Execution History table includes the following attributes:

Attributes	Description
Backup Job Information	Details of the backup job you created.
Backup End Time	The time when the backup job ended.
Completion Status	The current completion status of the backup job.
Mark for Archival	Select to archive backup job execution history after 90 days.

For more information, see [Operational Technology Backup Management data model](#).

Key reference structures

Use the following key references are important guidelines when creating Operational Technology (OT) records:

- When creating computer or server records for OT Devices that are running on computers or servers, see the following topics:
 - [Computer \[cmdb_ci_computer\] class](#)
 - [Server \[cmdb_ci_server\] class](#)
- The Manufacturer and Model ID attributes are reference attributes to the Company [core_company] and Product Model [cmdb_model] tables respectively.
- The Owned By, Assigned To, Managed By, and Supported By attributes are reference attributes to the User [sys_user] table. The Change Group and Support Group attributes are reference attributes to the Group [sys_user_group] table.

Identification rules

The ServiceNow AI Platform contains a predefined identification rule for the Operational Technology (OT) classes. That identification rule has the following key identifier entries, listed in priority order:

1. Identifier entry that uses the identification specified in Serial Number [cmdb_serial_number] as the lookup table. The Serial Number table is a many-to-one reference from the serial number back to the server CI.

Optional conditions to filter the records during identification and reconciliation.

Note: For lookup identification rules, you must complete the mandatory conditions under **Advanced Configuration**.

2. Identifier entry that is specified in the Serial Number attribute in the CI.
3. Identifier entry that is specified in the Mac Address attribute in the Network Adapter table.

Optional conditions to filter the records during identification and reconciliation.

4. Identifier entry for the Name attribute.
 - If Serial Number and MAC Address are not available, then the Name (which is usually the system reported hostname) attribute is used.
 - If both Serial Number and Name are not available, and only MAC Address is available, use MAC Address as the name of the CI. Using the MAC Address as the name of the CI ensures that you don't create an empty CI.
5. Identifier entry that uses a product instance identifier (PID) on the CI.

Note: To learn more, see [CMDB Identification and Reconciliation \(IRE\)](#).

Reclassifying IT devices on your OT network

For Xanadu and later versions, you can create a Network Adapter identifier on the OT identification rule to reclassify IT devices on your OT network. Creating a Network Adapter identifier on the OT identification rule overwrites the Hardware identification rule's Network Adapter identifier entry. To do this, complete the following actions:

1. Navigate to **All > Configuration > CI Class Manager**.

2. Under **Hardware** in the **CI Classes** menu, select the **Operational Technology (OT)** class.
3. Under the **Class Info** menu, select **Identification Rule**.
4. Under **Identifier Entries**, select **Add**.
5. In the New Identifier Entry pop-up dialog box, choose **Use attributes from another table (Lookup table)** and select **Next**.
6. In the Create Identifier Entry dialog box, complete the following steps:
 - a. Select **Hardware table** from **Search On Table** dropdown list.
 - b. Enter **400** in the **Priority** field.
 - c. Select **Network Adapter** from the **LookUP Table** from dropdown list.
 - d. Select **Mac Address** from the **Criterion Attributes** dropdown list.
 - e. Select **Save**.

Note: IT devices on an OT network are not reclassified automatically. If you want to automatically reclassify IT devices to OT, you must upgrade to Xanadu or the latest ServiceNow AI Platform version with the latest CMDB CI Class Models.

Related entries

Related entries for OT identification rules allow OT Service Graph Connectors to create related records, such as OT Device records, backup records, and so on. The Operational Technology [cmdb_ci_ot] class rule contains the following related entries:

- Backup Storage Information [cmdb_backup_storage_information]
- External system metadata [cmdb_key_value_v2]
- OT Device [cmdb_ot_entity]
- Firmware Installation [cmdb_firmware_install]
- Key Value [cmdb_key_value]

- Software Instance [cmdb_software_instance]

The OT Control Module [cmdb_ci_ot_control_module] rule contains the following related entries:

- Backup Storage Information [cmdb_backup_storage_information]
- OT Device [cmdb_ot_entity]
- Firmware Installation [cmdb_firmware_install]
- Key Value [cmdb_key_value]
- Software Instance [cmdb_software_instance]

OT Device Network Connection data model

The OT Device Network Connection data model leverages the OT Device Network Connection [sn_ot_device_network_connection] table to identify source CIs and destination CIs. Identifying this data allows you to identify the device-to-device connections in your OT environment.

Note: The OT Device Network Connection table is available with the Industrial Core plugin. For more information about the Industrial Core plugin, see [Industrial Core plugin](#).

The OT Device Network Connection [sn_ot_device_network_connection] table references the CI relationships [cmdb_rel_ci] table with the Connects to :: Connected by relationship type. The device-to-device connections in CI relationships and the metadata of how the source and destination communicate are stored.

Note: This data is populated with the available OT integrations and cannot be manually created.

The OT Device Network Connection table is populated using data from the following CMDB tables.

Allocated IP Address [cmdb_ci_allocated_ip_address]

The following attributes in the Allocated IP Address [cmdb_ci_allocated_ip_address] table are populated by collected data:

Attribute label	Attribute name
IP Address	ip_address
Managed Network	managed_network
Is Broadcast	is_broadcast
Is Conflict	is_conflict
Is DHCP	is_dhcp
Is DNS	is_dns
Is Managed	is_managed
Is Reserved	is_reserved
Name	name

IP Network Subnet [cmdb_ci_ip_network_subnet]

The following attributes in the IP Network Subnet [cmdb_ci_ip_network_subnet] table are populated by collected data:

Attribute label	Attribute name
Name	name
Parent Pool	parent_pool
CIDR	cidr
Managed Network	managed_network

Managed Network [cmdb_ci_managed_network]

The following attribute in the Managed Network [cmdb_ci_managed_network] table is populated by collected data:

Attribute label	Attribute name
Name	name

Attribute label	Attribute name
Contained by CI	contained_by_ci

The OT Device Network Connection table also uses the following Identification Reconciliation Engine (IRE) rule. The IRE rule on the OT Device Network Connection table can help Service Graph Connectors align with a standardized way to create connection records.

- Identifier name: OT Device Network Connection
- Applies to: sn_ot_device_network_connection
- Independent: true
- Identifier entries:
 - Source IP Address
 - Destination IP Address
 - Destination Port
 - Source Managed Network
 - Destination Managed Network

Deprecated classes

CMDB CI Class Models: Release 1.30 soft deprecated (not removed, but marked as follows) with the release of the OT class model:

Class	Description
Human Machine Interface [cmdb_ci_hmi]	OT Devices moved to cmdb_ci_ot to support broader use cases. Use cmdb_ci_ot_hmi instead.
Manufacturing Device [cmdb_ci_manufacturing]	OT Devices moved to cmdb_ci_ot to support broader use cases. Use cmdb_ci_ot as the base class or other generic child classes as appropriate.

Class	Description
Programmable Logic Controller [cmdb_ci_plc]	OT Devices moved to cmdb_ci_ot to support broader use cases. Use cmdb_ci_ot_plc instead.

A script has been provided to migrate records from these classes to the new class tables. The admin role is required to perform the following tasks:

1. Navigate to **System Definition > Script Includes**
2. Find the record named **OTDevicesMigrationUtils**.
3. Navigate to **Definition > Script Includes > .**
4. In the background window, copy and paste the provided script.
5. Select the appropriate scope and run the script.

Troubleshooting

The following are some troubleshooting tips:

Problem	Suggested resolution
Unable to see OT Device menu items	Ensure that the logged in user has been assigned the appropriate roles. To learn more, see the preceding Roles section.
Error creating or updating an OT Device record	Allow only one OT Device record (cmdb_ot_entity) per CI.

Request apps on the Store

Visit the [ServiceNow Store](#) website to view all the available apps and for information about submitting requests to the store. For cumulative release notes information for all released apps, see the [ServiceNow Store version history release notes](#).

The [CMDB CI Class Models](#) adds class models that extend the CMDB class hierarchy, including:

- Class descriptions

- Identification rules
- Identifier entries
- Dependent relationships, if applicable.

Related concepts

- [CMDB schema model](#)

Red Hat Virtualization (RHV) extension classes

The CMDB CI Class Models store app adds or updates classes for Red Hat Virtualization (RHV) OLVM or RHV.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

Starting with version 1.28.0, Discovery and Service Mapping Patterns supports both OLVM and RHV discovery. For more information, see [Oracle Linux Virtualization Manager and Red Hat Virtualization discovery](#). Note that the RHV patterns, tables, and related items have been renamed to oVirt. For the full list of name changes, see the Zurich release [ITOM Visibility release notes](#).

See the [full release notes for all CMDB CI Class Models](#).

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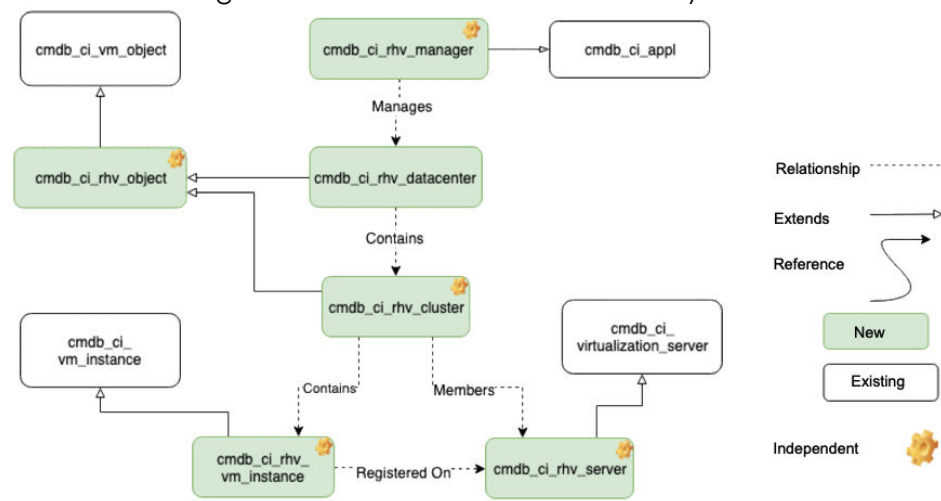
Red Hat Virtualization (RHV)

Red Hat Virtualization (RHV) is a virtualization product that is based on the Kernel-based Virtual Machine (KVM) hypervisor. RHV uses the SPICE protocol and Virtual Desktop Server Manager (VDSM) with a RHEL-

based centralized management server. The RHV solution is based on two primary software components: Red Hat Virtualization Manager (RHV-M) and Red Hat Virtualization Hypervisors or hosts: Red Hat Enterprise Linux or RHV Host (RHV-H).

Note: The Discovery application uses patterns to find RHV components. Discovering some of these resources requires updating the Discovery and Service Mapping Patterns application from the ServiceNow Store. For more information, see [Oracle Linux Virtualization Manager and Red Hat Virtualization discovery](#)

RHV classes integrated with the CMDB class hierarchy



Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. See the class columns table for further details about the columns added for each class.

CMDB CI Class Models: Release 1.8.0 adds the following classes for RHV. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Extends	Description
RHV LDC	Logical Datacenter	RHV logical datacenter.

Class	Extends	Description
[cmdb_ci_rhv_idc]	[cmdb_ci_logical_datacenter]	
RHV Datacenter [cmdb_ci_rhv_datacenter] This class is deleted.	N/A	N/A

CMDB CI Class Models: Release 1.6.0 adds the following classes for RHV.

Class	Extends	Description
RHV Server [cmdb_ci_rhv_server]	Virtualization Server [cmdb_ci_virtualization_server]	The RHV virtualization host.
RHV Object [cmdb_ci_rhv_object]	Virtual Machine Object [cmdb_ci_vm_object]	A base class for other classes to derive from.
RHV Manager [cmdb_ci_rhv_manager]	Application [cmdb_ci_appl]	RHV Manager instance.
RHV Cluster [cmdb_ci_rhv_cluster]	RHV Object [cmdb_ci_rhv_object]	RHV cluster.
RHV Datacenter	RHV Object [cmdb_ci_rhv_object]	RHV datacenter.

Class	Extends	Description
[cmdb_ci_rhv_datacenter]		
Note: This class is being deleted in the CMDB CI Class Models 1.6.0 release.		
RHV Virtual Machine Instance [cmdb_ci_rhv_vm_instance]	Virtual Machine Instance [cmdb_ci_vm_instance]	RHV virtual machine instance.

Class columns

CMDB CI Class Models: Release 1.6.0 adds the following columns to the respective classes.

RHV Server [cmdb_ci_rhv_server] class

Added columns	Description
url	URL used to access the object.

RHV Object [cmdb_ci_rhv_object] class

Added columns	Description
href_id	Href ID.
manager_id	Manager ID.
url	URL to access the object (used in child classes).

RHV Manager [cmdb_ci_rhv_manager] class

Added columns	Description
url	URL of the manager.
product_name	Product name.

RHV Cluster [cmdb_ci_rhv_cluster] class

Added columns	Description
cpu_type	CPU type.
cpu_architecture	CPU architecture.
kvm	Enabled state of kernel same-page merging (KSM) memory policy.
fencing_policy	Fencing policy.
memory_overcommitment	Amount of over-commitment memory allowed on the cluster.
transparent_huge_pages	Transparent huge memory pages policy.
ballooning	Memory ballooning for guests.
compatibility_version	Compatibility version.

RHV Virtual Machine Instance [cmdb_ci_rhv_vm_instance] class

Added columns	Description
mgmt_url	Management URL.
cpu_architecture	CPU architecture.
delete_protected	Delete protected (true/false).

Added columns	Description
ha_priority	HA priority.
high_availability	High availability (true/false).
memory_policy_guaranteed	Amount of memory guaranteed (MB).
memory_policy_max_mb	Maximum memory in the dynamic memory allocation policy.
multi_queues	Multi queues.
placement_policy	Placement policy.
cpu_sockets	Number of CPU sockets.
stateless	Stateless (true/false).
storage_error_resume_behaviour	Behavior of a virtual machine that is paused due to storage I/O error. For examples, AUTO_RESUME, LEAVE_PAUSED, and KILL.
start_time	Start time.
stop_time	Stop time.
threads	Number of threads.
time_zone	Time zone.
usb	USB enabled state.
run_once	Run once.
type	RHV type.

The following class was deleted in the CMDB CI Class Models 1.6.0 release.

RHV Datacenter [cmdb_ci_rhv_datacenter] class

Added columns	Description
quota_mode	Quota mode policy.

Related concepts

- [CMDB schema model](#)

Related topics

- [Oracle Linux Virtualization Manager and Red Hat Virtualization discovery](#)

Transport Layer Security (TLS) extension classes

The CMDB CI Class Models store app adds or updates a class for TLS certificates.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

Request apps on the Store

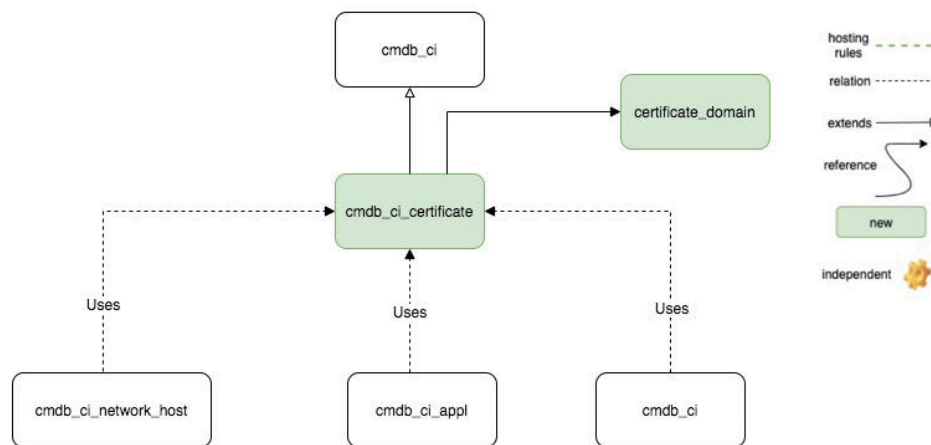
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Transport Layer Security (TLS)

TLS is a cryptographic protocol designed to provide communications security over a computer network. The TLS protocol provides privacy and data integrity between communicating computer applications. Once

the client and the server have agreed to use TLS, they negotiate a stateful connection by using a handshaking procedure. The server usually provides identification in the form of a digital certificate. The certificate contains the server name, the trusted certificate authority (CA) that vouches for the authenticity of the certificate, and the server's public encryption key. The client confirms the validity of the certificate before proceeding. When the handshake is completed, a secured connection is established.

TLS certificate classes integrated with the CMDB class hierarchy



Scoped apps certification class

The scoped apps certification class supports TLS certificates. With this class you can proactively manage certificates by keeping stakeholders informed about any impending expiries. Use this extension class to ensure that certificates are monitored and renewed before they expire, to prevent severe outage of production systems.

Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. See the class columns table for further details about the columns added for each class.

The CMDB CI Class Models: Release 1.4.0 updates the following class:

Class	Extends	Description
Unique Certificate [cmdb_ci_certificate]	Configuration Item [cmdb_ci]	A public key certificate in X.509 standard format.

The CMDB CI Class Models store app changes the Unique Certificate [cmdb_ci_certificate] class as follows:

- The assigned_to attribute now depends on the assignment_group attribute so that users in the assigned_to attribute are filtered based on the specified assignment_group.
- The [Certificate Inventory and Management](#) store app populates the Unique Certificate [cmdb_ci_certificate] table. The list view for that class does not have a **New** button and you can no longer add new records to the table. This is because there are certain fields that are extracted from binary encoded parameters in the certificate which users may not be able to provide. Also, certificates have to be discovered rather uploaded.
- You can no longer add or delete attachments in the Certificate file attribute.

The CMDB CI Class Models: Release 1.3.0 adds the following classes. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Extends	Description
Unique Certificate [cmdb_ci_certificate]	Configuration Item [cmdb_ci]	N/A
Certificate Domain [certificate_domain]	N/A	Fully qualified domain name.

Class columns

CMDB CI Class Models: Release 1.4.0 adds the following columns to the respective classes.

Unique Certificate [cmdb_ci_certificate] class

Added columns	Description
Certificate file	Certificate in an encoded form.
Fingerprint	Hash value of the certificate.
Fingerprint algorithm	Algorithm used to hash the certificate.
Is certificate authority	Indicates whether a certificate is a Certificate Authority (CA) or not.
Is selfsigned	Indicates whether the certificate is self-signed or not.
Issuer	Entity that has signed and issued the certificate. Reference: Unique Certificate [cmdb_ci_certificate]
Issuer common name	Common name of the issuer.
Issuer distinguished name	Distinguished name of the issuer.
Key size	Size of the key used by the signing algorithm. Choices: • Create priority 1 tasks • Create priority 3 tasks • Do not create renewal tasks
Renewal tracking	Indicates whether to create any priority 1 or priority 3 tasks for the expiring certificates.

Added columns	Description
Root issuer	Root entity that has signed and issued the intermediate certificate. Choices: • External • Internal Reference: Unique Certificate [cmdb_ci_certificate]
Service type	Indicates whether the certificate is used for external or internal services.
Signature algorithm	The cryptographic algorithm used to sign the certificate. Choices: • Issued • Installed • Revoked • Retired
State	Lifecycle states of the certificate.
Subject alternative name	List of fully qualified domain names secured by the certificate. Reference: Certificate Domain [certificate_domain]
Subject common name	Identifies the hostname/domain associated with the certificate.

Added columns	Description
Subject country	Subject's two letter country code.
Subject distinguished name	Identifying information of the subject.
Subject email	Subject's email.
Subject locality	Subject's locality.
Subject organization	Subject's organization.
Subject organizational unit	Subject's organizational unit.
Subject state	Subject's state.
Valid from	Validity start period of the certificate.
Valid to	Validity end period of the certificate.
Version	X.509 version of the certificate.

Certificate Domain [certificate_domain] class

Added columns	Description
Domain	Fully qualified domain name.

CMDB CI Class Models: Release 1.3.0 adds no columns.

Related concepts

- [CMDB schema model](#)

VMware NSX load balancer extension classes

The CMDB CI Class Models store app adds or updates classes for VMware NSX load balancers.

The CMDB CI Class Models app adds class models that extend the CMDB class hierarchy, including class descriptions, identification rules, identifier entries, and, if applicable, dependent relationships. You can use the added classes just like any other CMDB class. Applications such as Discovery and Service Mapping Patterns can use the class extensions to populate CIs and discover technologies and software.

See the [full release notes for all CMDB CI Class Models](#).

Request apps on the Store

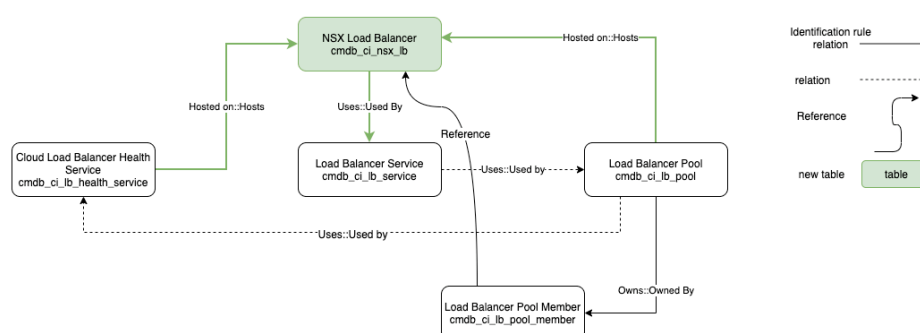
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VMware NSX load balancer

NSX is a network virtualization solution offered by VMware. Among the virtual resources included in the NSX solution are virtual LANs (VLANs), virtual load balancers, virtual routers, switches, and firewalls.

Discovery uses the [VMware NSX Advanced load balancer discovery](#) pattern to find VMware NSX load balancers and their components: Listeners, pools, pool members, and health services.

VMware NSX load balancer classes integrated with the CMDB class hierarchy



Classes

This section lists the classes that the CMDB CI Class Models store app adds or updates. See the class columns table for further details about the columns added for each class.

CMDB CI Class Models: Release 1.10.0 adds the following classes for the VMware NSX load balancer. For the list of classes in the base system, including classes that this app might extend, see [CMDB tables descriptions](#).

Class	Extends	Description
NSX Load Balancer [cmdb_ci_nsx_lb]	Load Balancer [cmdb_ci_lb]	The table containing the NSX Load Balancer resources.

Class columns

The [VMware NSX Advanced load balancer discovery](#) pattern introduces one new table with one identification rule and entry. The table uses only the columns inherited from its parent.

Related concepts

- [CMDB schema model](#)